

Active Learning

Agenda

- What is Active Learning
- Methods for selecting texts for labeling
- Active Learning details

Difficulty in getting data

Any dataset for supervised learning must be annotated **manually**.

Text annotation is a complex and expensive task
(it can cost hundreds of thousands of dollars depending on the task).

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How long does it take you to identify named entities in a text?

Normunds Viltinis has headed the Anti-Corruption Bureau since 2009. During his tenure, as Telegraf notes, KNAB officers carried out a number of high-profile arrests, including the former head of Latvenergo, Kārlis Miķelsons, and the former head of the Customs Criminal Investigation Department, Vladimirs Vashkevich. One of KNAB's most recent high-profile cases was a series of searches at companies associated with prominent Latvian oligarchs.

Difficulty in getting data

Sometimes labelling requires a highly qualified assessors.

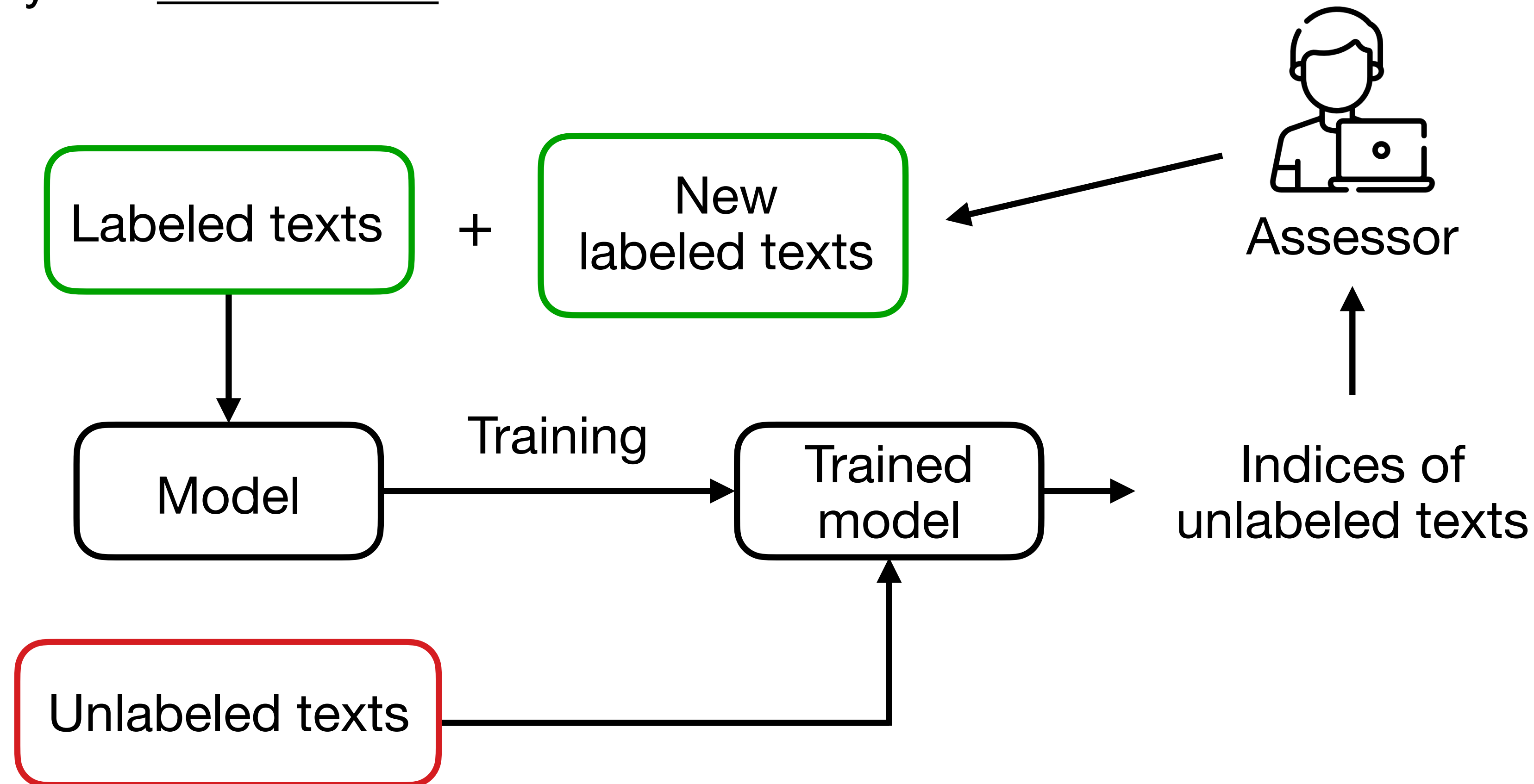
- Bilingual skills for translation tasks
- Linguistic education for text markup

What if you need a lot of labeled data and your budget is limited?

Active Learning

A paradigm for training models with a limited labeling budget.

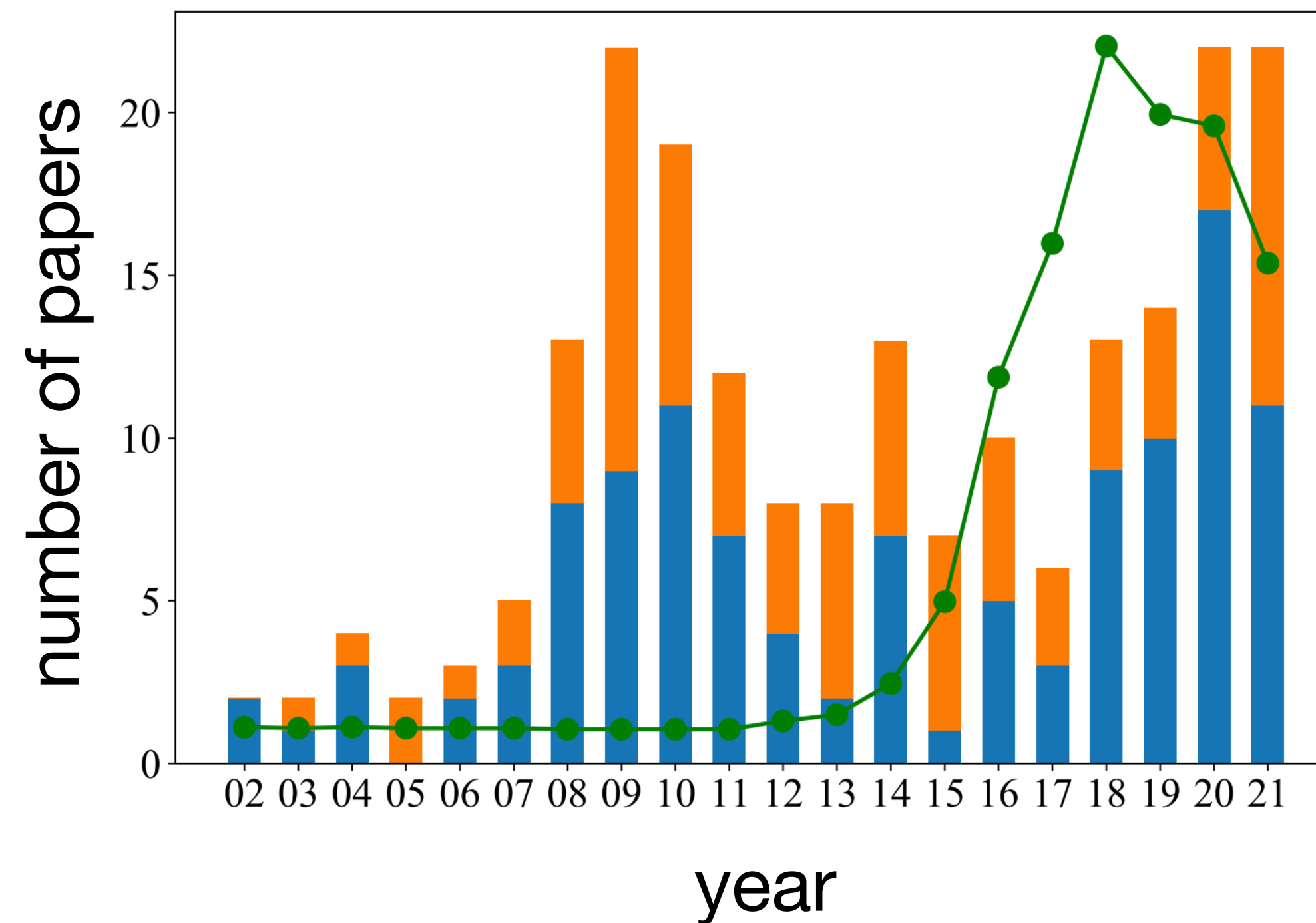
Idea: Alternating model training with additional data labeling, labeling only the most useful texts



Active Learning

Active Learning lost its popularity for a time, but regained it with the introduction of Transformer in 2017.

Dynamics of articles about AL by year

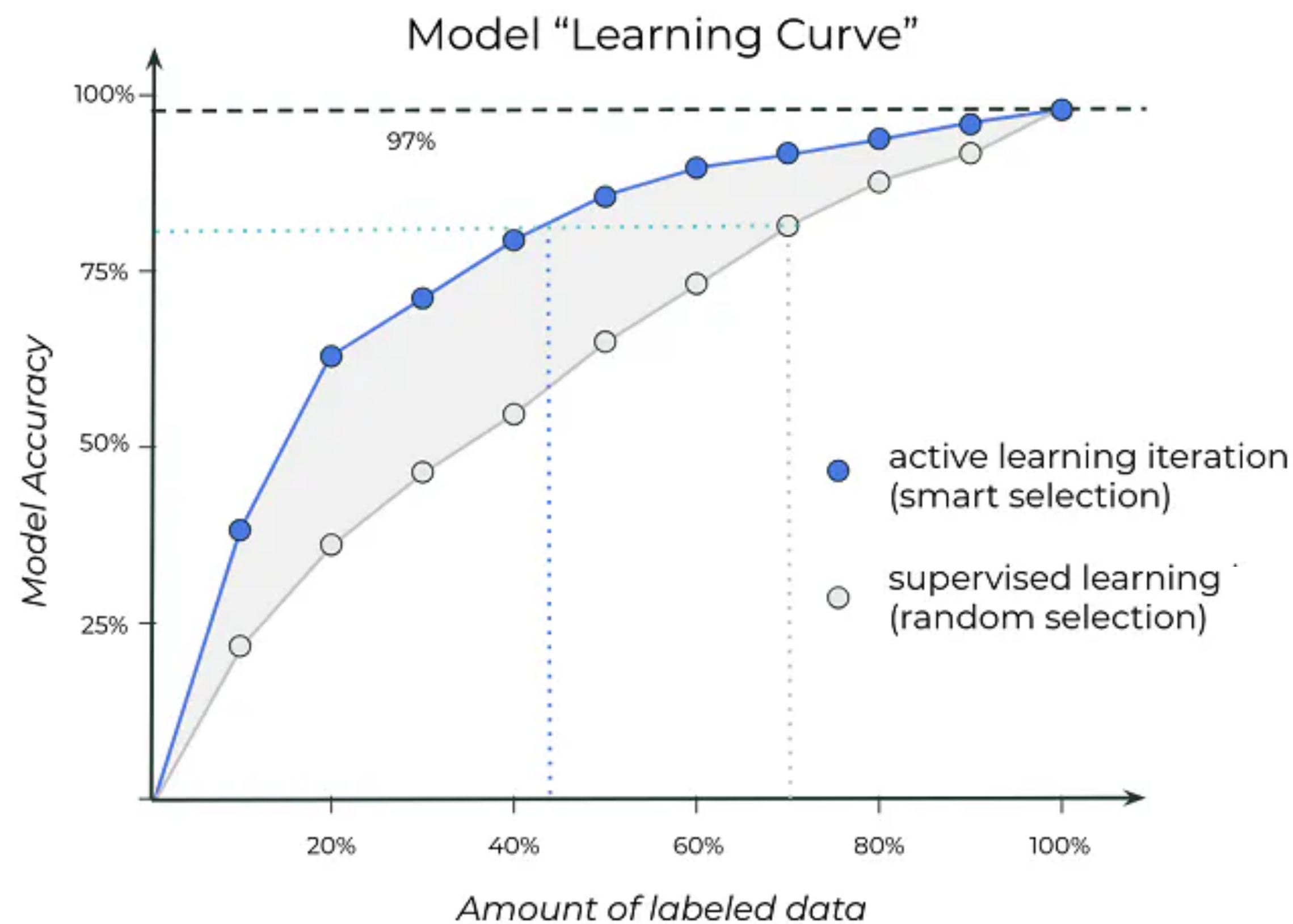


Active Learning: main questions

- How to choose texts for annotation?
- How to tune the model?
- When you should stop?
- How to speed up the training and annotation cycle?

How to chose texts for annotation?

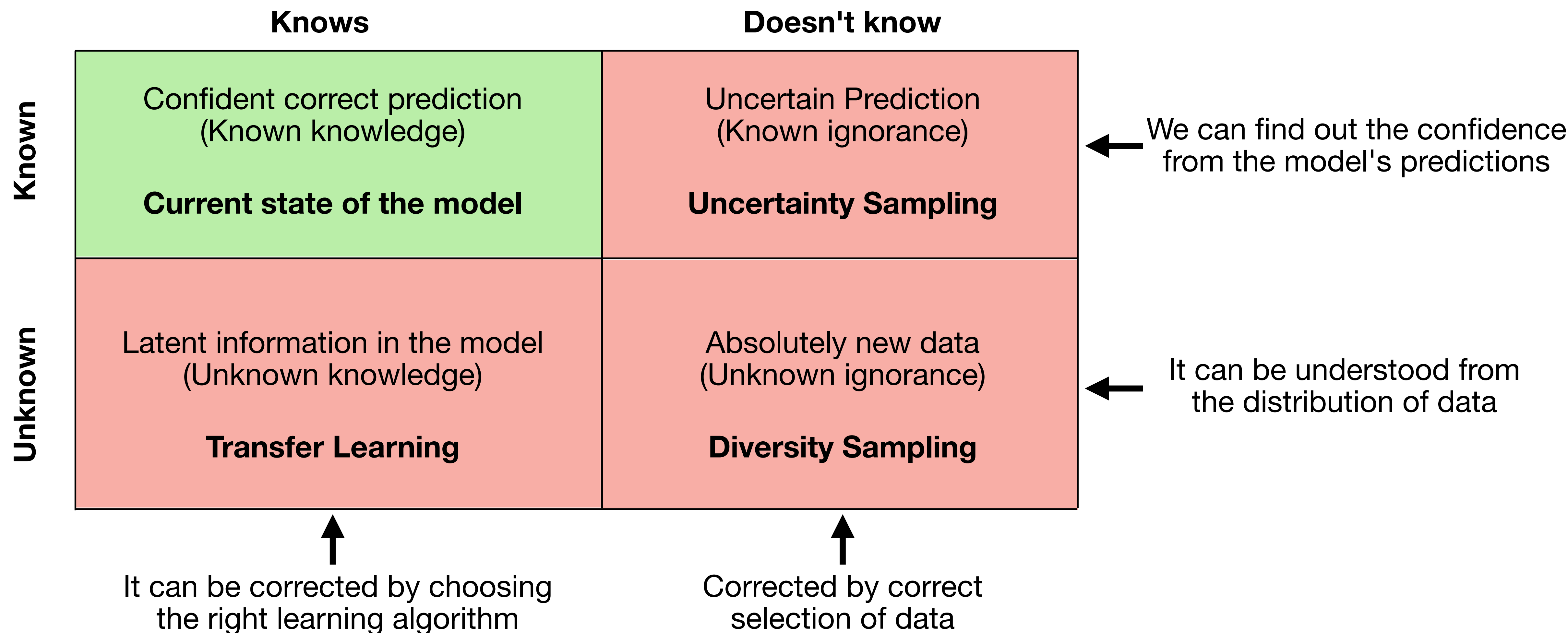
- Chose randomly
- Choose the texts that are most important to train on



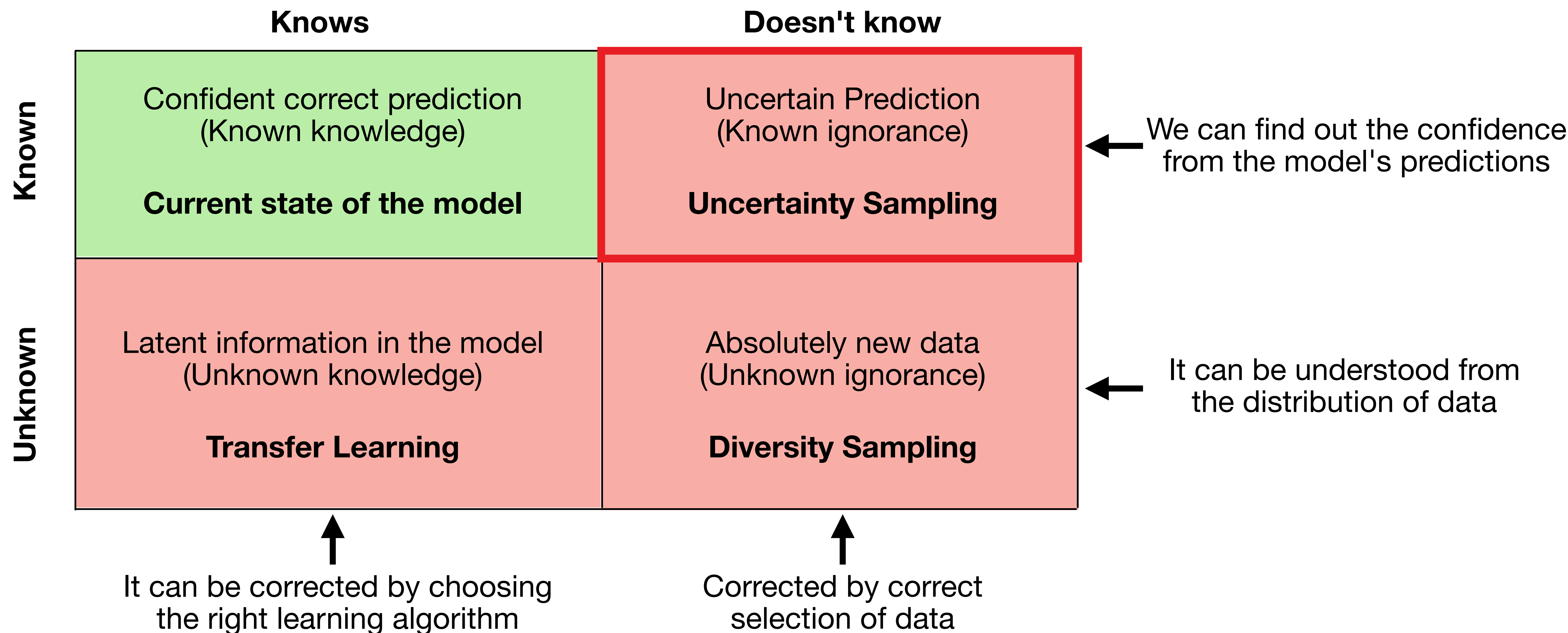
Square of model knowledge

	Knows	Doesn't know
Known	Confident correct prediction (Known knowledge) Current state of the model	Uncertain Prediction (Known ignorance) Uncertainty Sampling
Unknown	Latent information in the model (Unknown knowledge) Transfer Learning	Absolutely new data (Unknown ignorance) Diversity Sampling

Square of model knowledge



Square of model knowledge

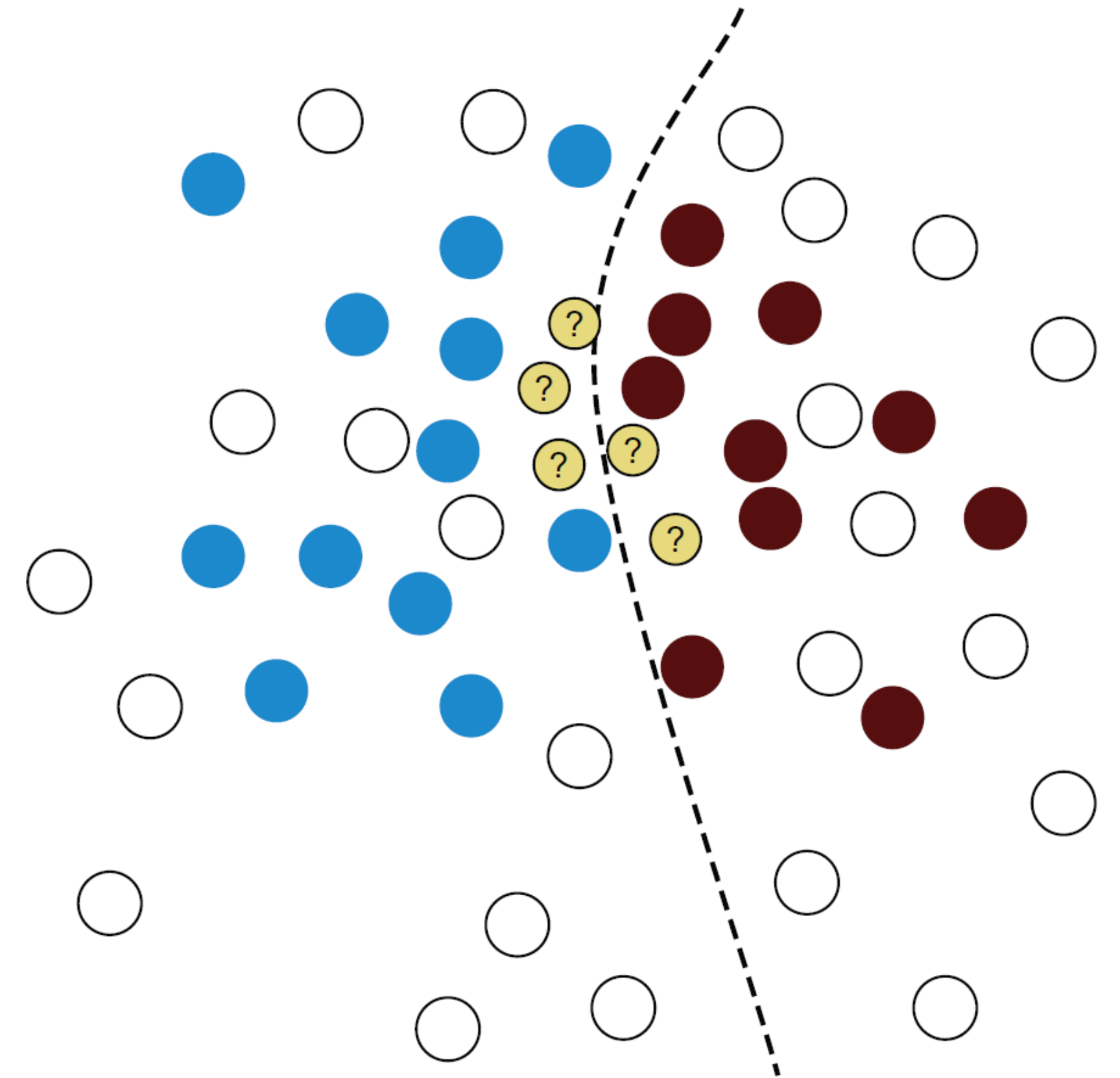


Uncertainty Sampling

- It's important to train the model on those objects it's unsure about.
- Such objects are located near the separating curve.

Algorithm:

- Calculate the confidence for all unlabeled objects.
- Submit the n least confident objects for labeling.
- Retrain the model with these objects.



How to evaluate confidence?

We'll consider a number of methods:

- Least Confident
 - Margin of Confidence
 - Ratio of Confidence
 - Label Entropy
-
- Query by Committee

Least Confident

We estimate confidence based on the probability of the most probable class.

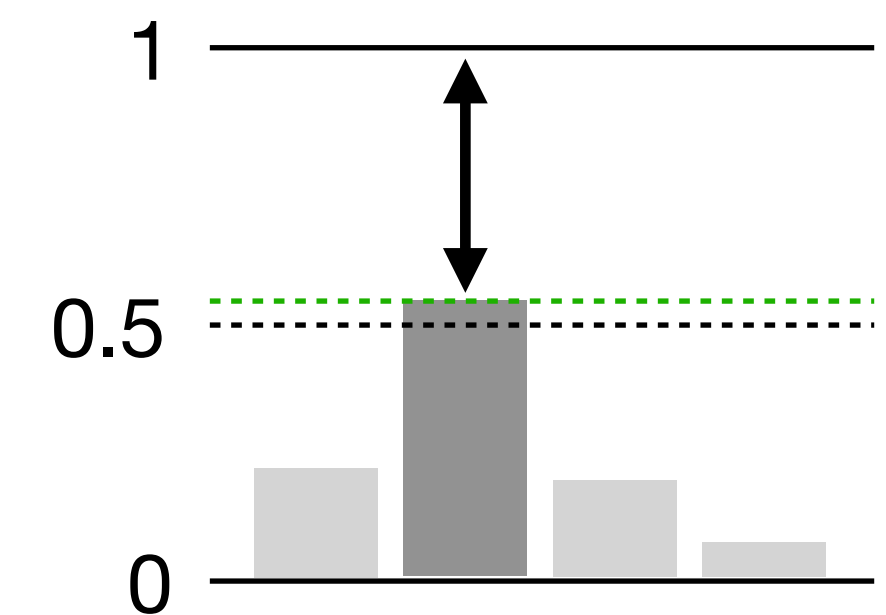
$$\text{score}(x) = 1 - \mathbb{P}(\hat{y} | x),$$

where $\hat{y}$ is the most probable class

class A	class B	class C	class D
0.43	0.07	0.21	0.29

$$\text{score}(x) = \mathbb{P}(A | x) = 0.43$$

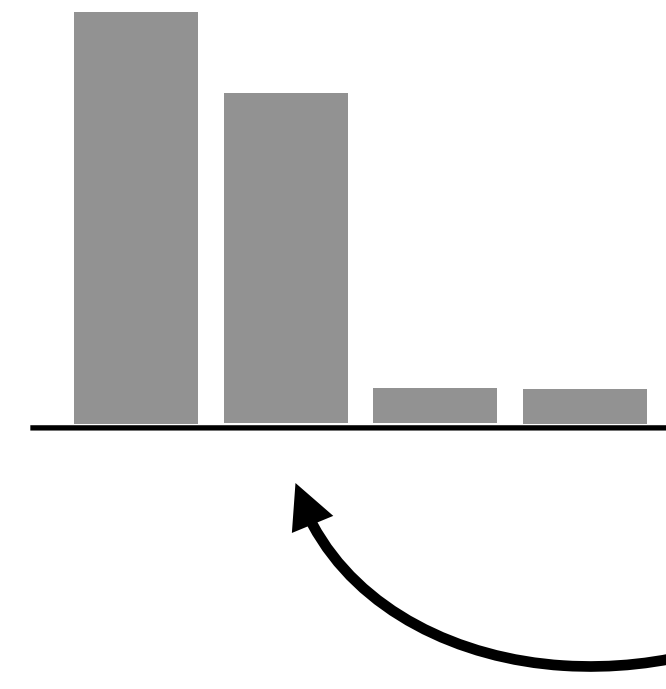
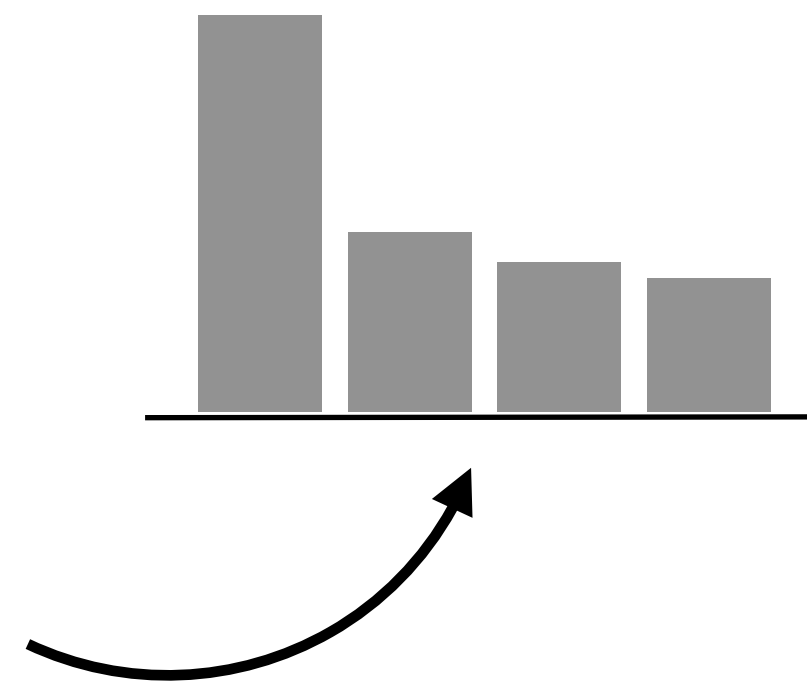
Objects with the highest score are selected for labeling.



Least Confident: limitations

- The method works well for binary classification.
- It performs worse for multi-class classification because it doesn't take into account the probabilities of other classes.

This distribution looks better, but we will choose it



this one has a higher maximum probability

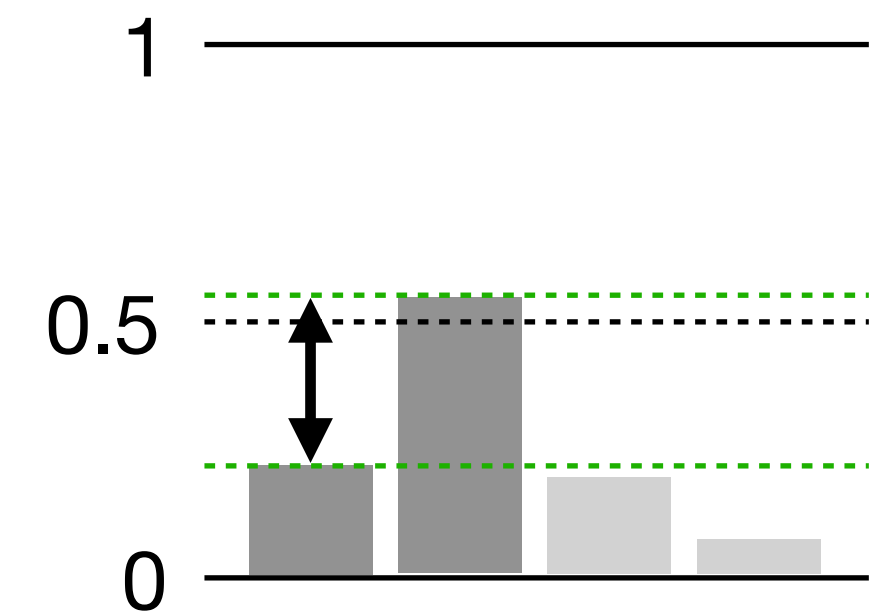
Margin of Confidence

- Calculate the difference between the two most probable classes

$$\textit{score}(x) = 1 - (\mathbb{P}(\hat{y}_1 | x) - \mathbb{P}(\hat{y}_2 | x)),$$

where $\hat{y}_1$ and $\hat{y}_2$ are the first and second most probable classes

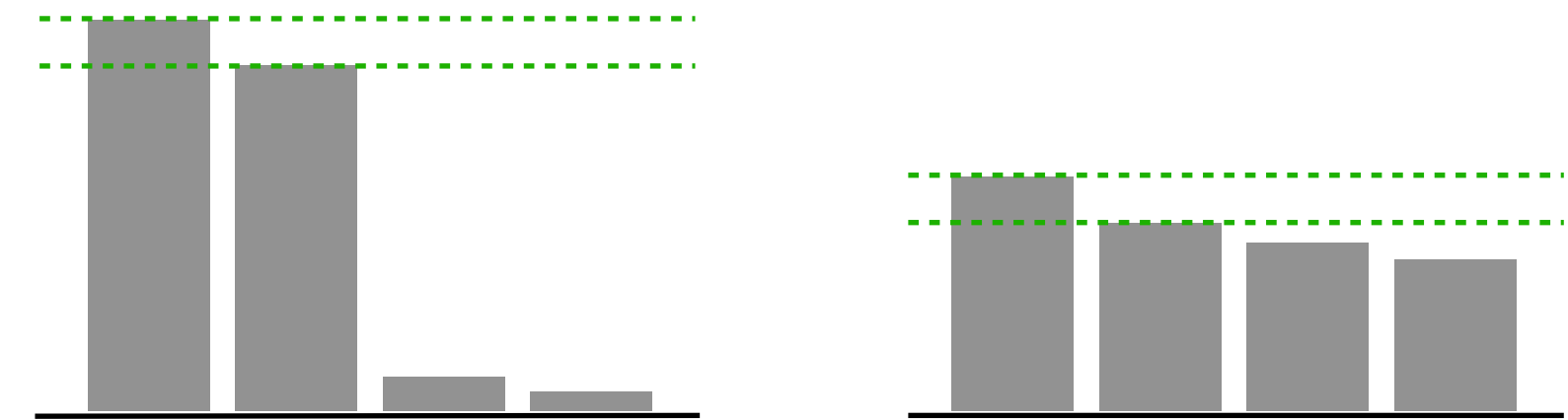
This method is used most often.
But it doesn't always work correctly.



Margin of Confidence: limitations

- The score is independent of the other probabilities.
- When looking at the difference, we don't take into account the probability values themselves.

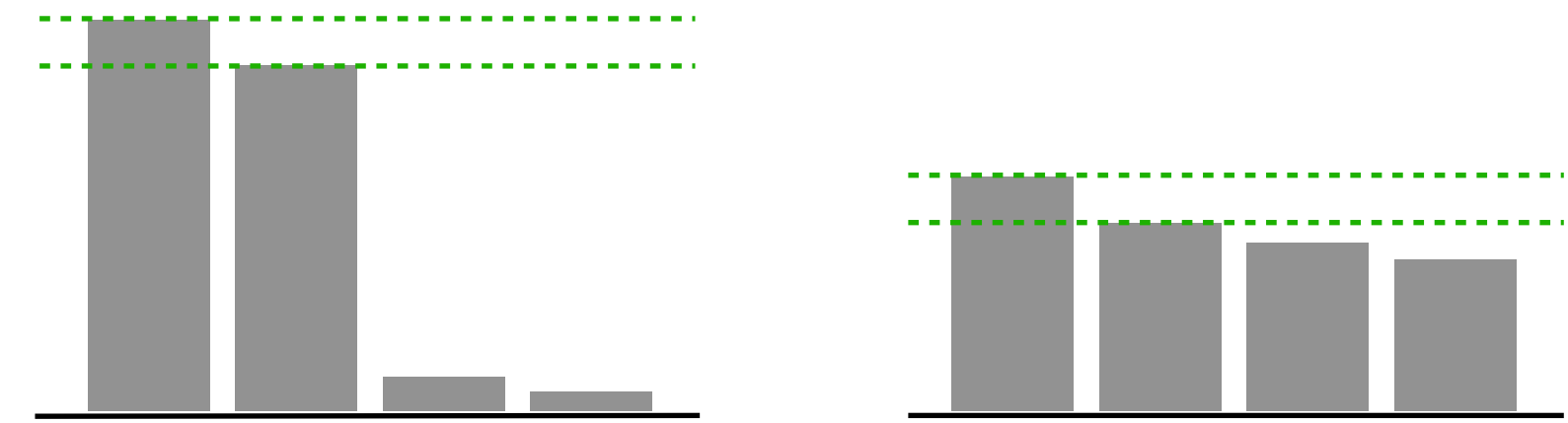
For these distributions, the difference in the probabilities of the two most probable classes is the same.



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- To correct this, you can adjust the temperature parameter in softmax.
- But it's better to use the Ratio of Confidence.

Ratio of Confidence

- Probabilities are calculated using softmax

$$\textit{softmax}(z_i) = \frac{e^{z_i}}{\sum_{k=1}^K e^{z_k}}$$

- softmax ignores bias of values

$$\textit{softmax}([1,2,3]) = \textit{softmax}([101,102,103])$$

Ratio of Confidence

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- Calculate the ration between the two most probable classes

$$\text{score}(x) = \frac{\mathbb{P}(\hat{y}_2 | x)}{\mathbb{P}(\hat{y}_1 | x)},$$

where $\hat{y}_1$ and $\hat{y}_2$ are the first and second most probable classes

Ratio of Confidence

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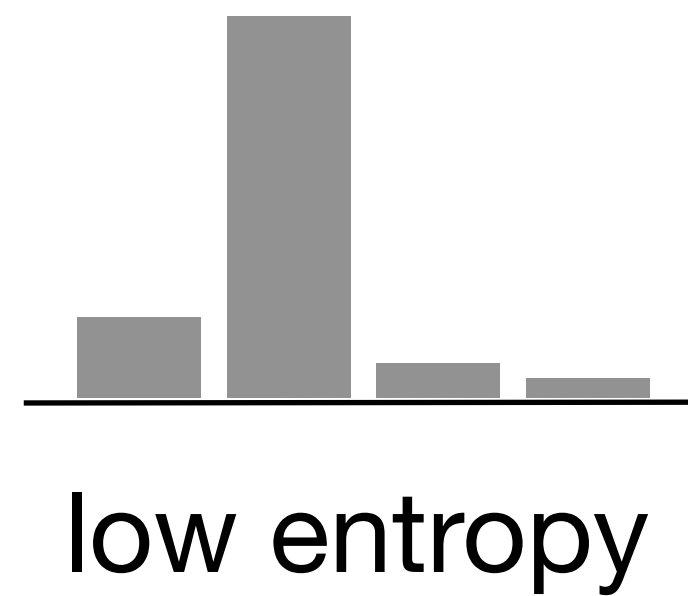
$$\text{score}(x) = \frac{\mathbb{P}(\hat{y}_2 | x)}{\mathbb{P}(\hat{y}_1 | x)} = \frac{e^{\hat{z}_2}}{\sum_{k=1}^K e^{z_k}} \cdot \frac{\sum_{k=1}^K e^{z_k}}{e^{\hat{z}_1}} = e^{\hat{z}_2 - \hat{z}_1}$$

- The score depends only on the difference; there's no need to adjust the temperature.
- The method still doesn't take into account the probabilities of the remaining classes.

Label Entropy

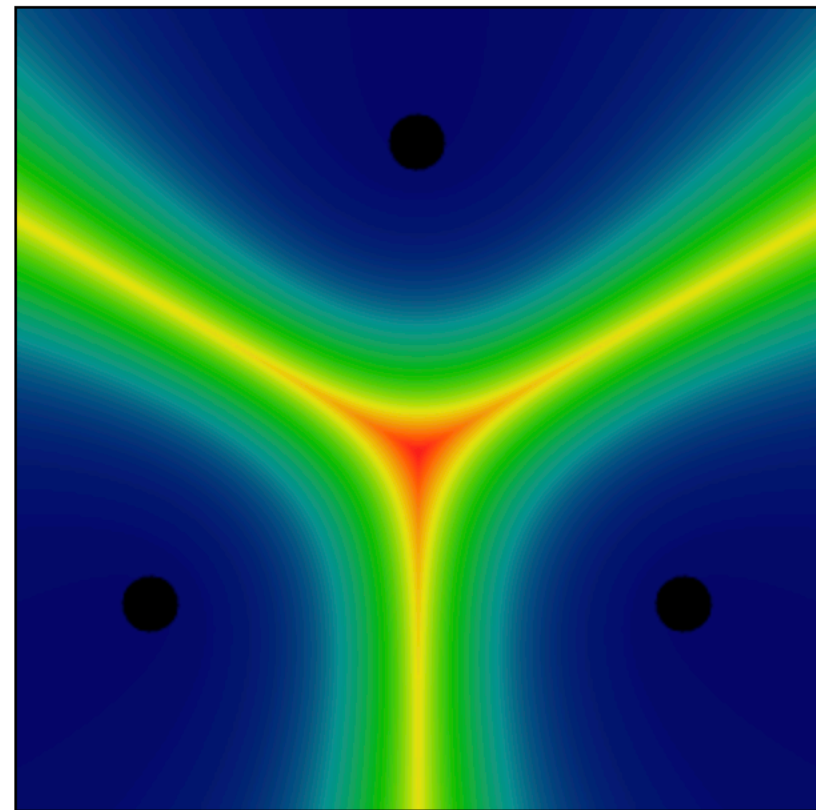
Entropy is a measure of the uncertainty of a probability distribution

$$score(x) = - \sum_{k=1}^K \mathbb{P}(y_k | x) \log \mathbb{P}(y_k | x)$$

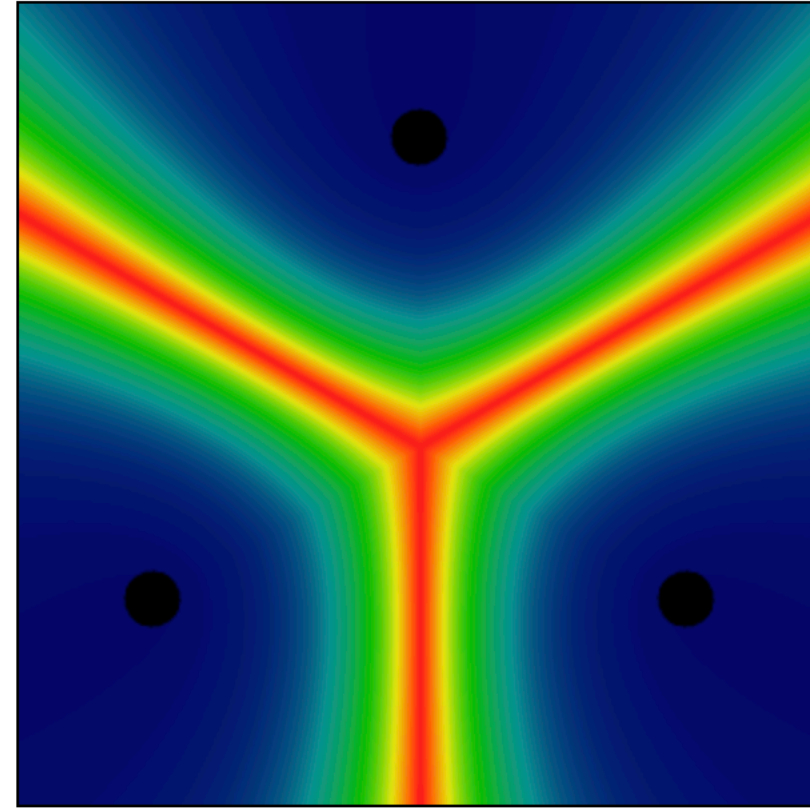


Visual comparison of methods

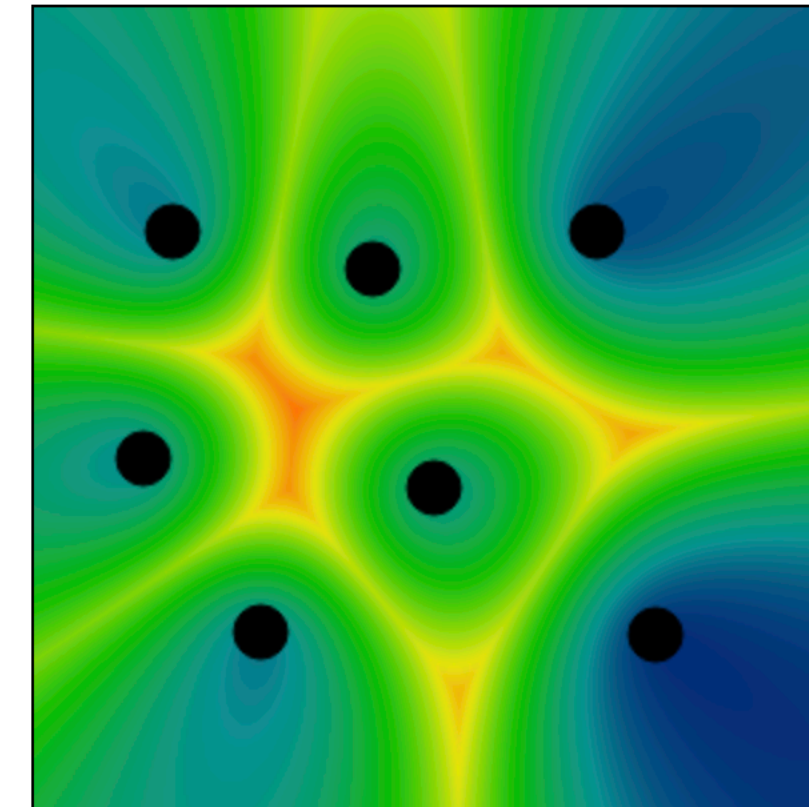
Least Confident



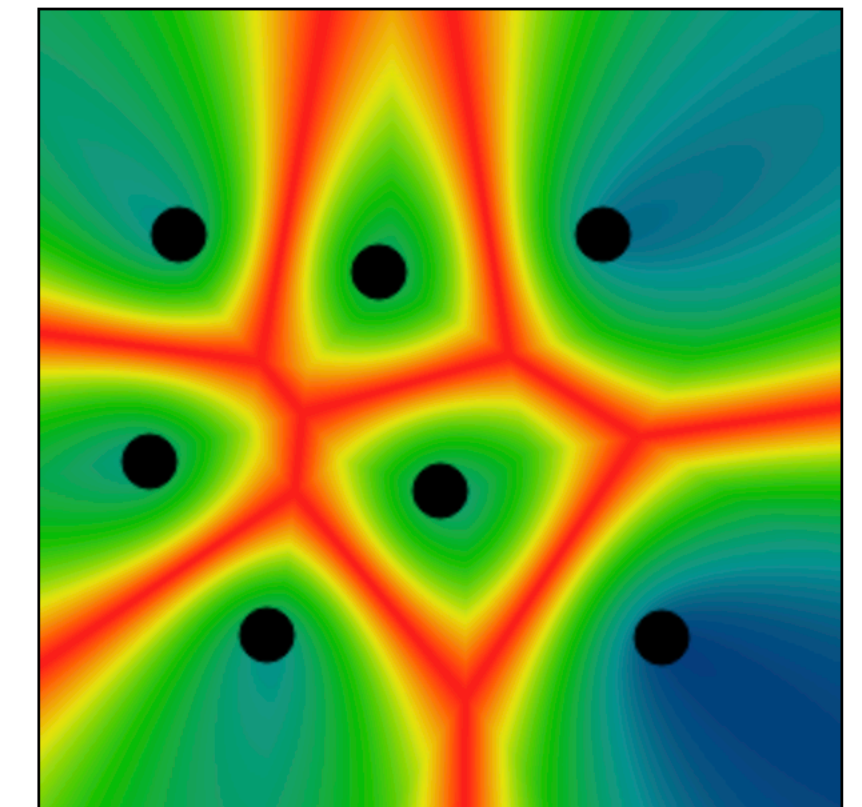
Margin of Confidence



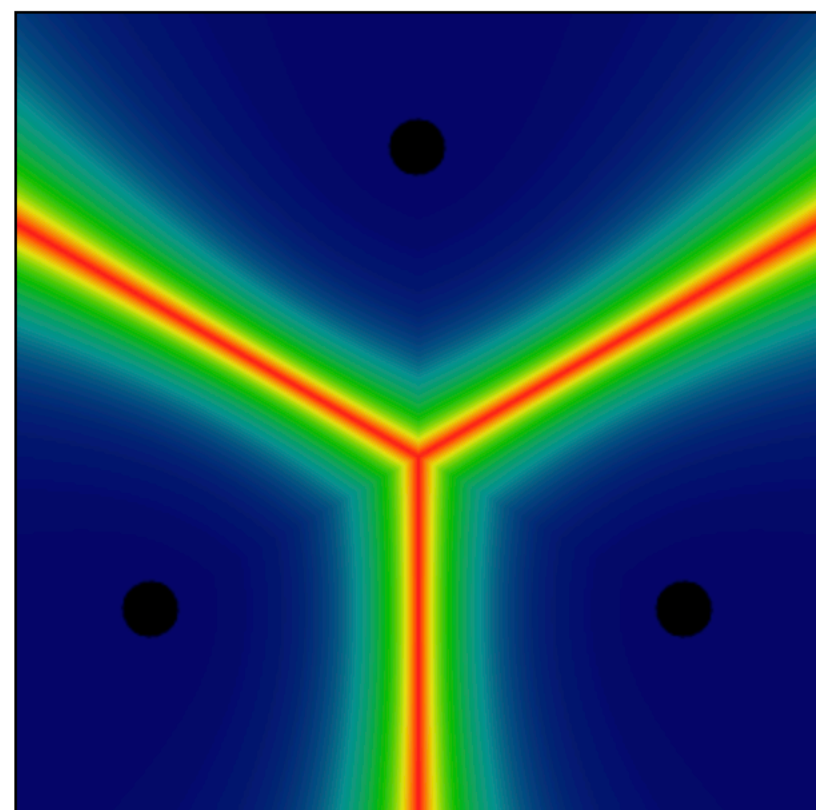
Least Confident



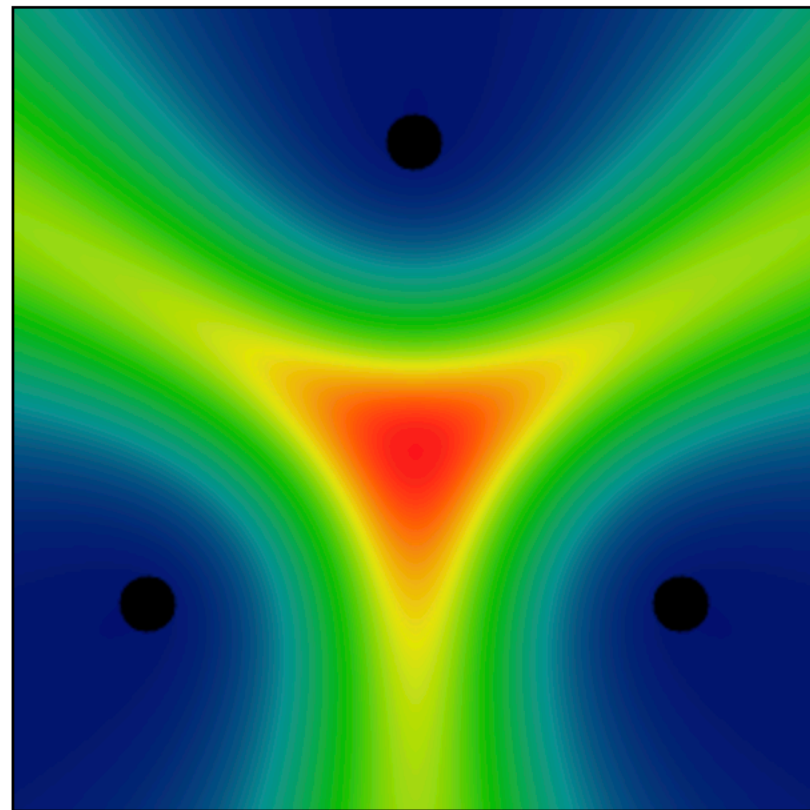
Margin of Confidence



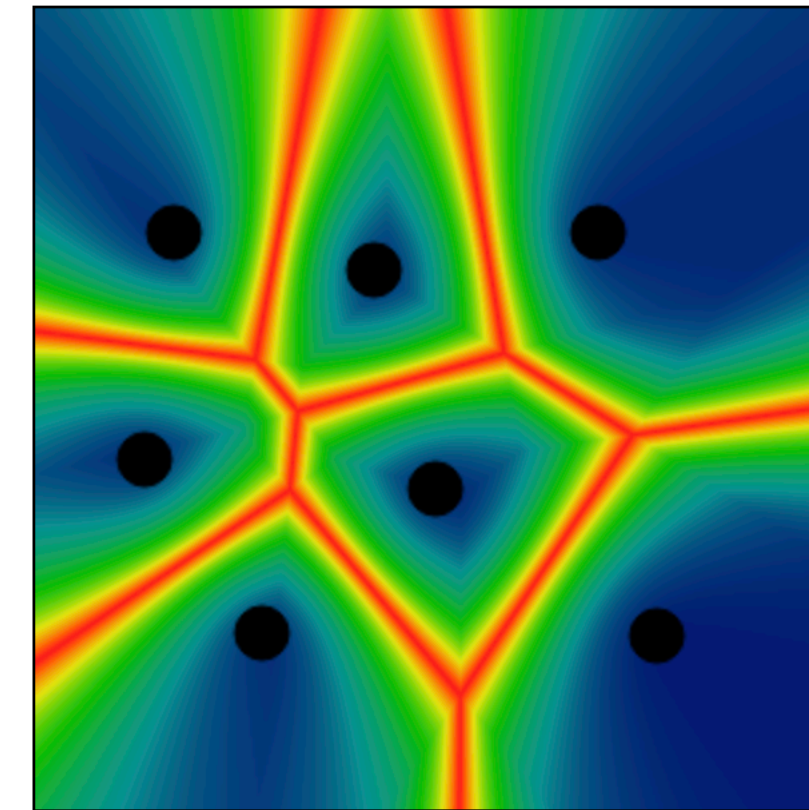
Ratio of Confidence



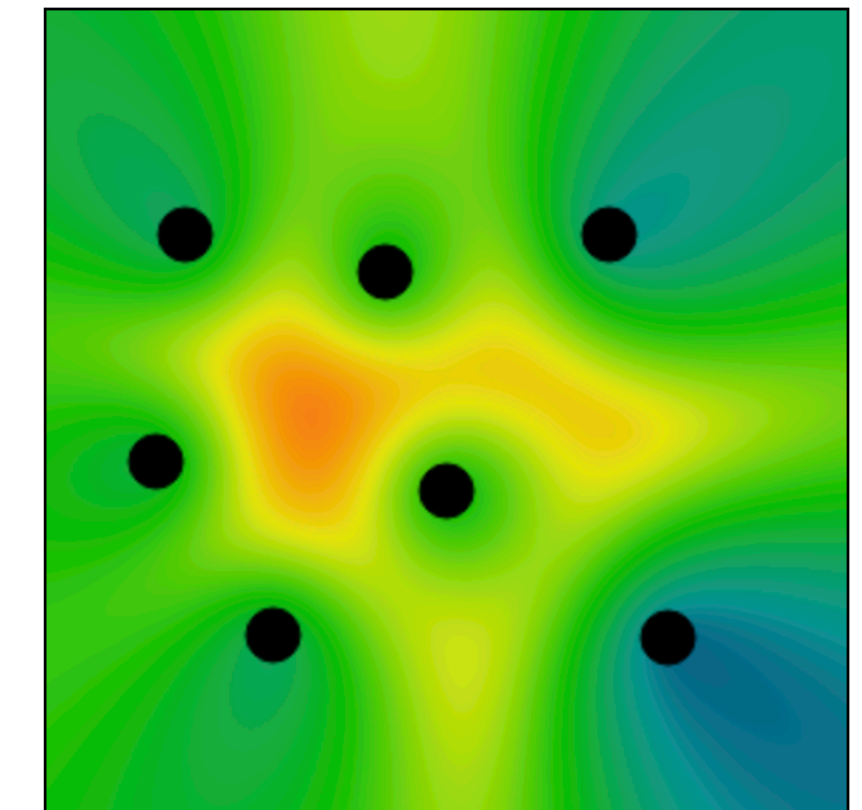
Label Entropy



Ratio of Confidence



Label Entropy



Query by Committee

- We train an ensemble of m models and look at the consistency of their predictions.
- Entropy is most often used.

$$score(x) = - \sum_{k=1}^K \frac{V(y_k)}{m} \log \frac{V(y_k)}{m},$$

where $V(y_k)$ is the number of models that predicted the class y_k .

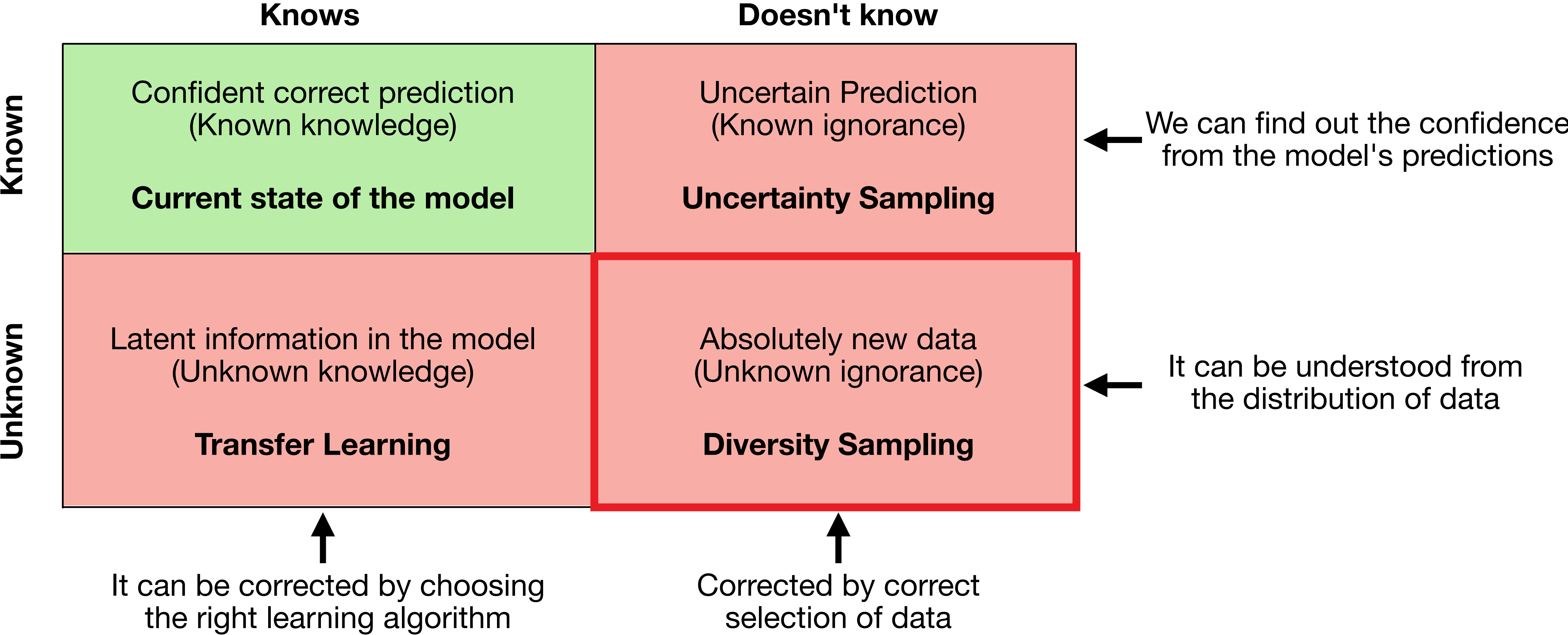
You can look at probability distributions and estimate confidence based on

- KL divergence
- Minimum probability of the most probable class
- Average confidence of the models

Query by Committee: limitations

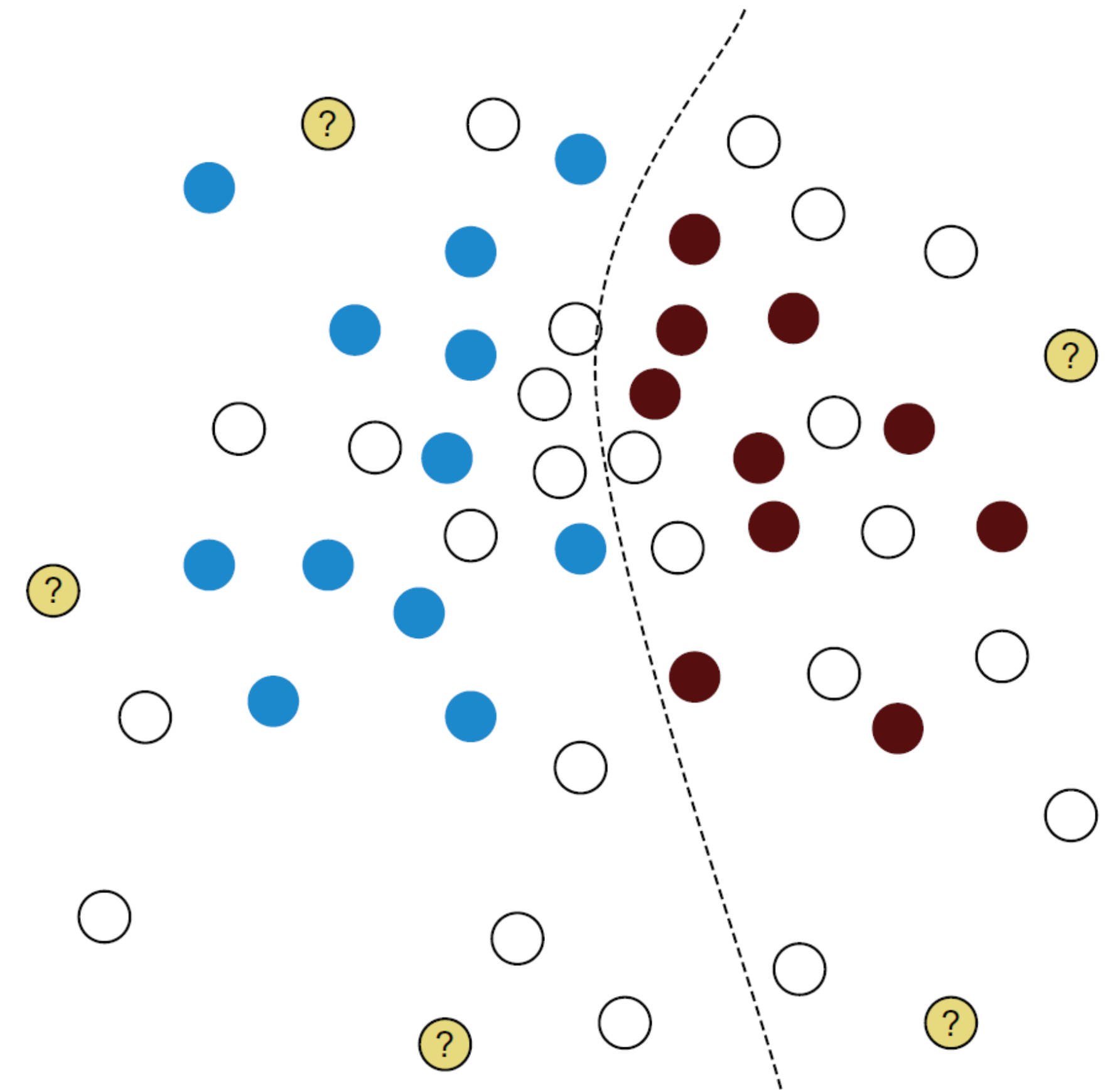
- Training m models is computationally inefficient.
- Models can have high correlations.
 - You can train different models on different subsets of data (bagging).
 - You can train models from different families: neural networks, TF-IDF, etc.
- Objects with the highest scores are often outliers.
- It is important to carefully select the sampling method for objects for labeling.

Square of model knowledge



Diversity Sampling

- Texts in real data are often biased:
 - By language
 - By source
 - By class (negative reviews outnumber positive ones)
- There are many reasons for this bias
 - More developed countries generate more data
 - Popular sources have the resources to create content
- We want the model to be able to work with all types of data



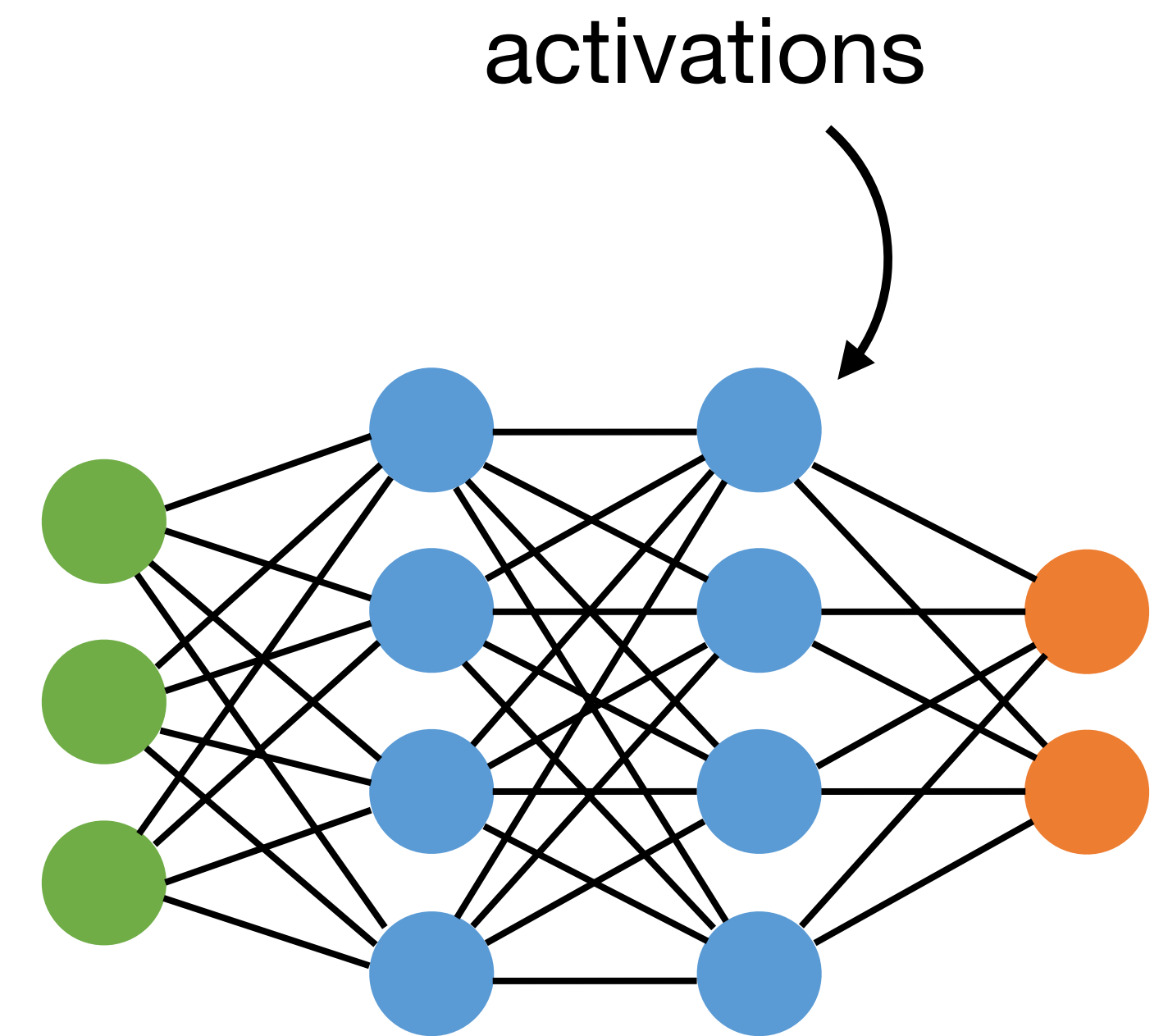
Diversity Sampling

Diversity can be increased in various ways.

- Model-based sampling
- Cluster-based sampling
- Representative sampling

Model-based sampling

- Since models are neural networks, you can look at the activations of their layers.
- If the input text activates a neuron, then the model is likely familiar with the text.
- If none of the neurons activate, then the model knows nothing about the text.



What does it mean to be activated?

- If the activation value is 1.34, is that a lot or a little?
- The validation sample (unlabeled) can be used to estimate the activation magnitude.
- Let a specific neuron receive values for different validation texts.

[2.52, 1.95, 1.18, 0.22, 0.12]

- Let's say 2.52 corresponds to 100% activation, and 0.12 corresponds to 0% activation.
- Then the new text with a value of 0.17 will lie exactly between 20% and 0%. It will be activated at 10%.

Neuronal aggregation

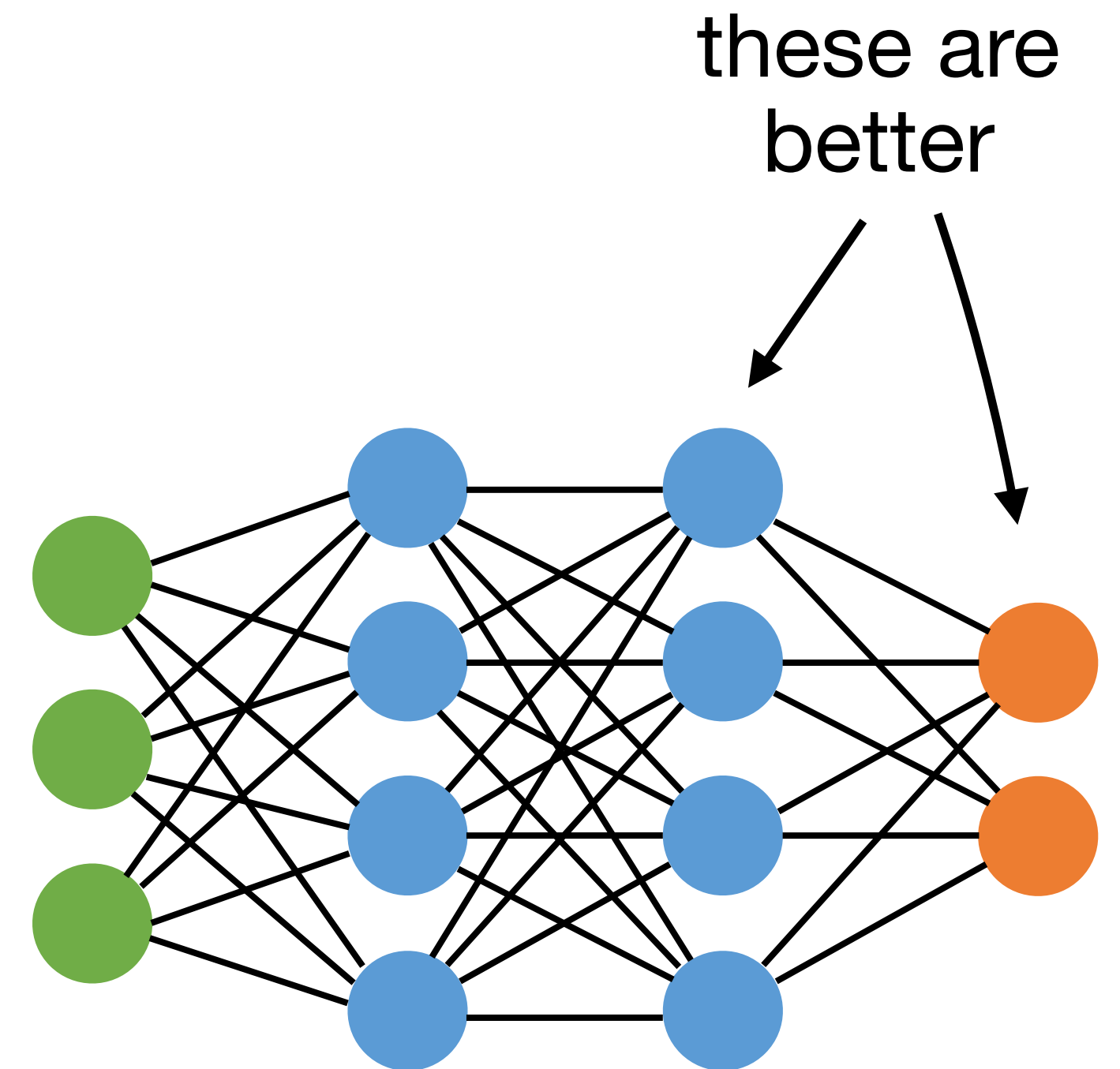
The most reliable way to aggregate ranks is averaging.

Let's say we have d output neurons. Let's calculate the activation level for each and average the values.

$$score(z) = \frac{1}{d} \sum_{i=1}^d rank(z_i)$$

Which layers' activations should you take?

- The deeper the layer, the more semantic information it extracts.
- Deeper layers are usually better.
- Multiple layers can be used.
- It's safer to extract layer outputs before activation!

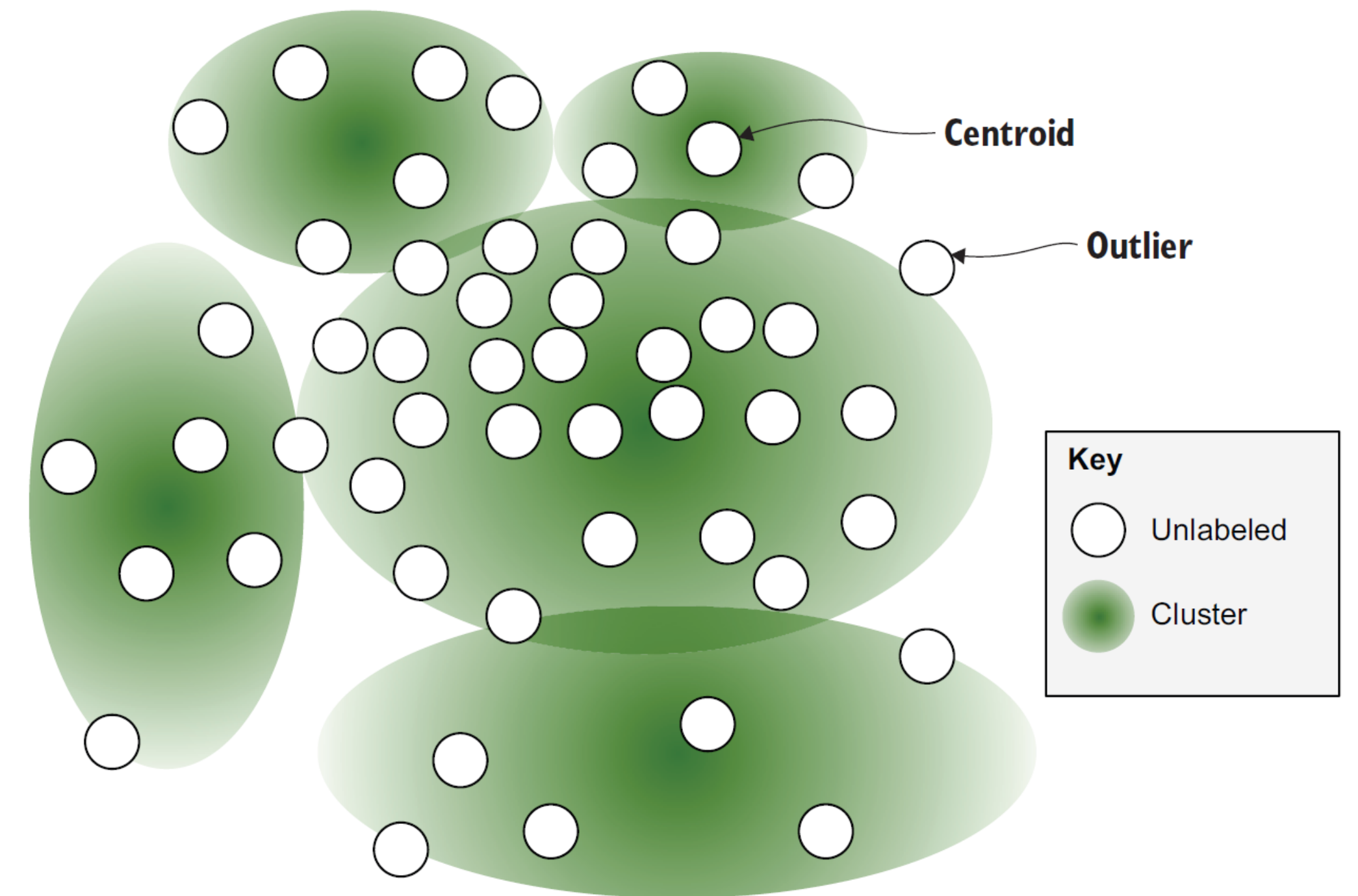


Limitations of the Model-based method

- The texts found may be very similar.
- The method performs poorly initially, when data is sparse.
- Due to statistical biases in the models, we may not be able to detect some outliers.

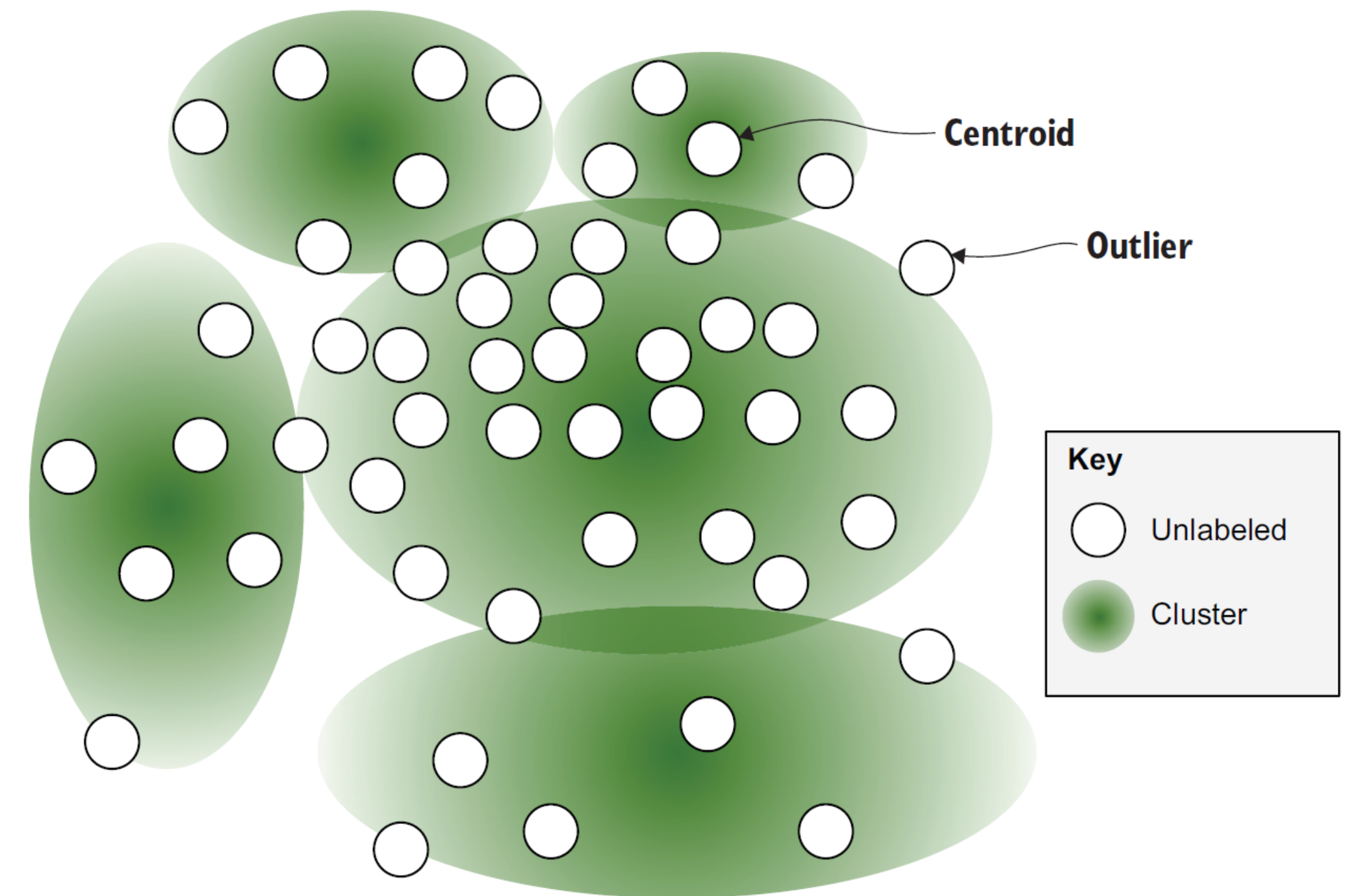
Cluster-based sampling

- We split the sample into multiple clusters
- We take several texts from each cluster



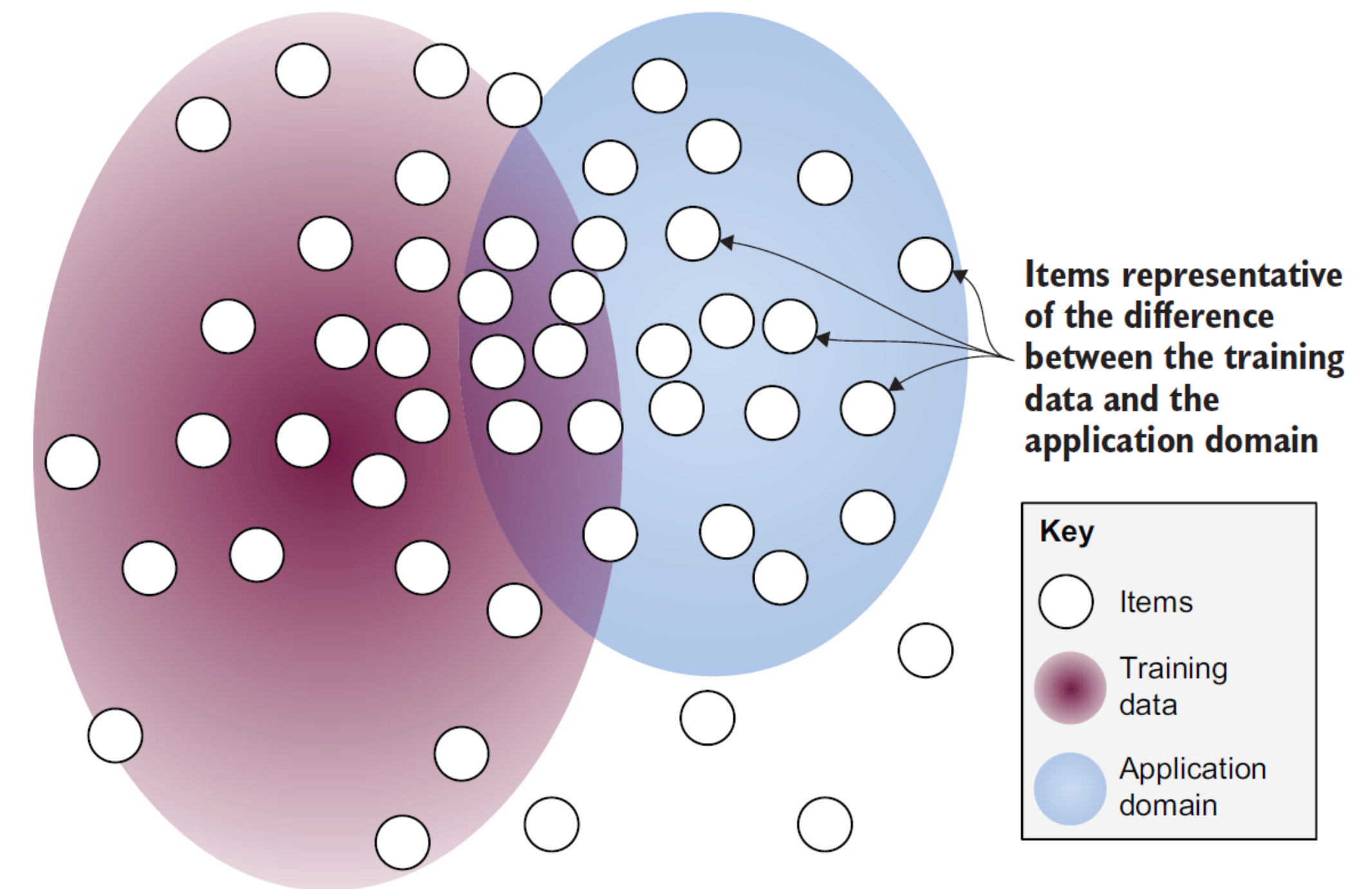
Cluster-based sampling

- We split the sample into multiple clusters
- We take several texts from each cluster
- When selecting texts, you can use a **random**, **centroid**, or **outlier**.
- Usually, it is better to use a little bit of everything.



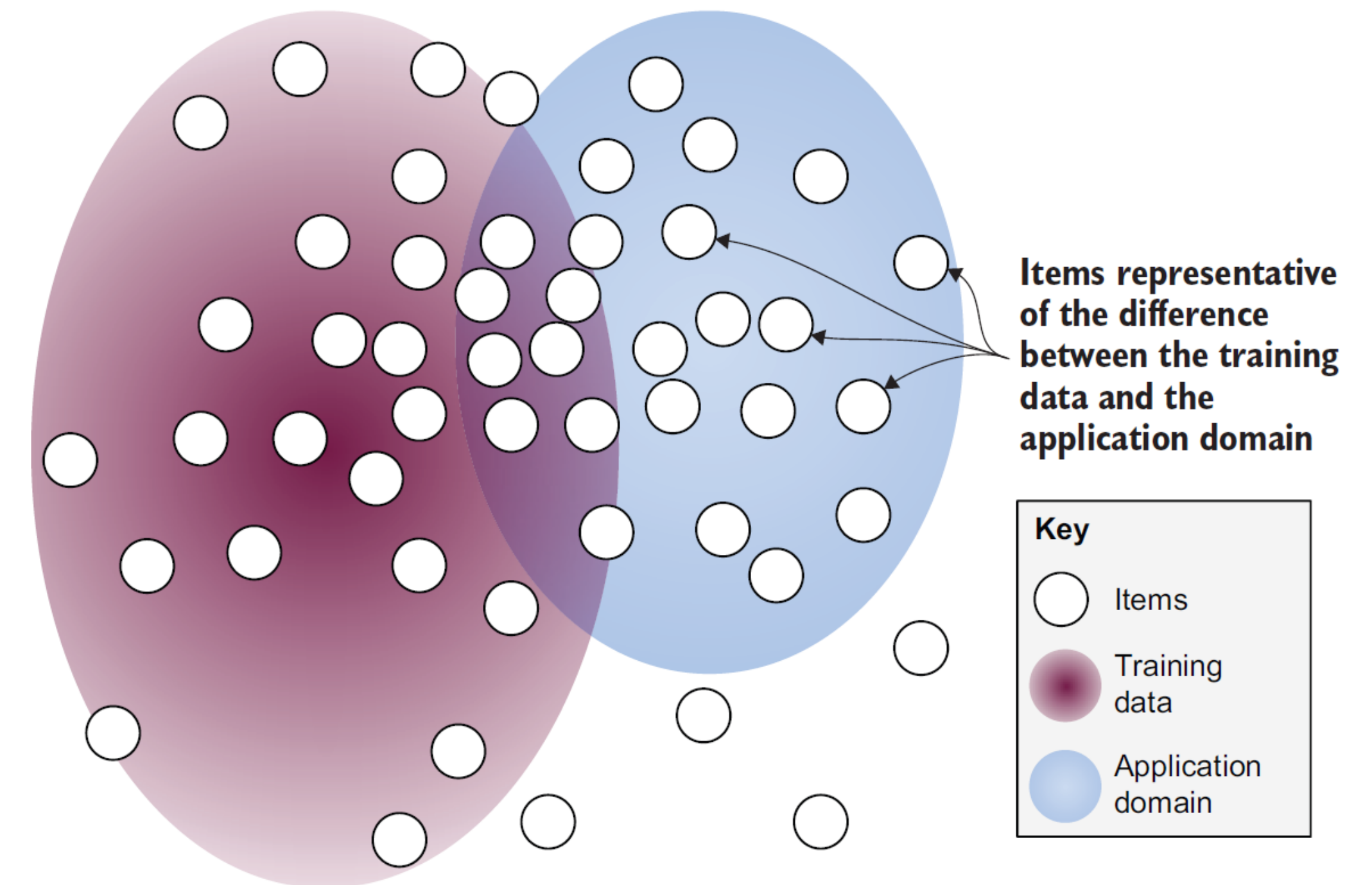
Representative sampling

- In real-world data, the distribution of unlabeled data often differs from the target domain.
- If we know the target distribution, we will select texts that are closest to it.
- To assess proximity, we can cluster both samples and compare the distance of the text to the nearest clusters.



Limitations of Representative sampling

- Representative sampling often leads to overfitting and low text diversity.
- It's best used in combination with Uncertainty Sampling.

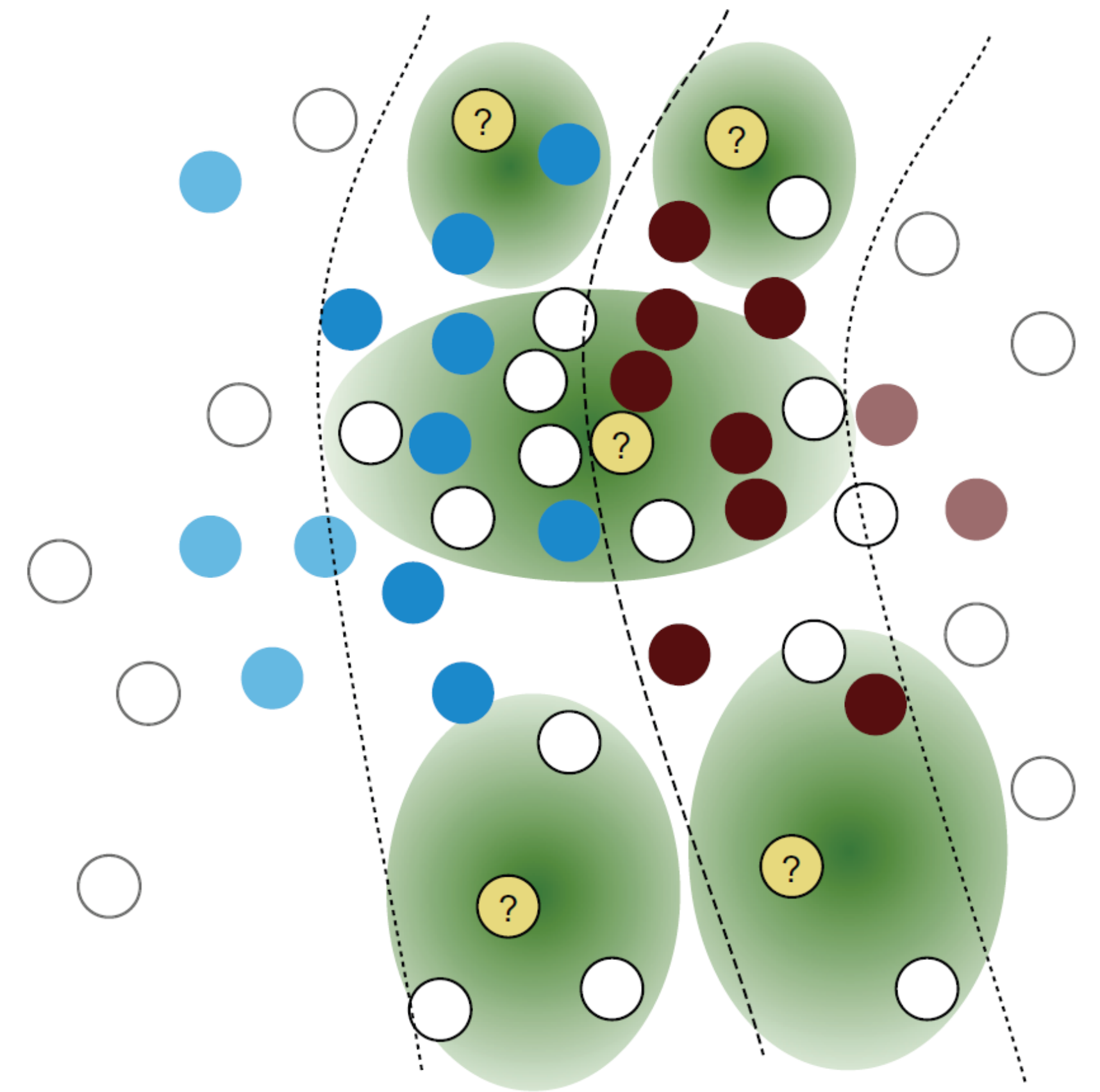


Combination of methods

- Uncertainty Sampling often leads to text similarities and poor model generalization.
- Diversity Sampling produces insufficiently complex examples.
- It's very logical to combine methods to address these shortcomings.
- In early cycles, Diversity Sampling is more important, while in later cycles, Uncertainty Sampling is more important.

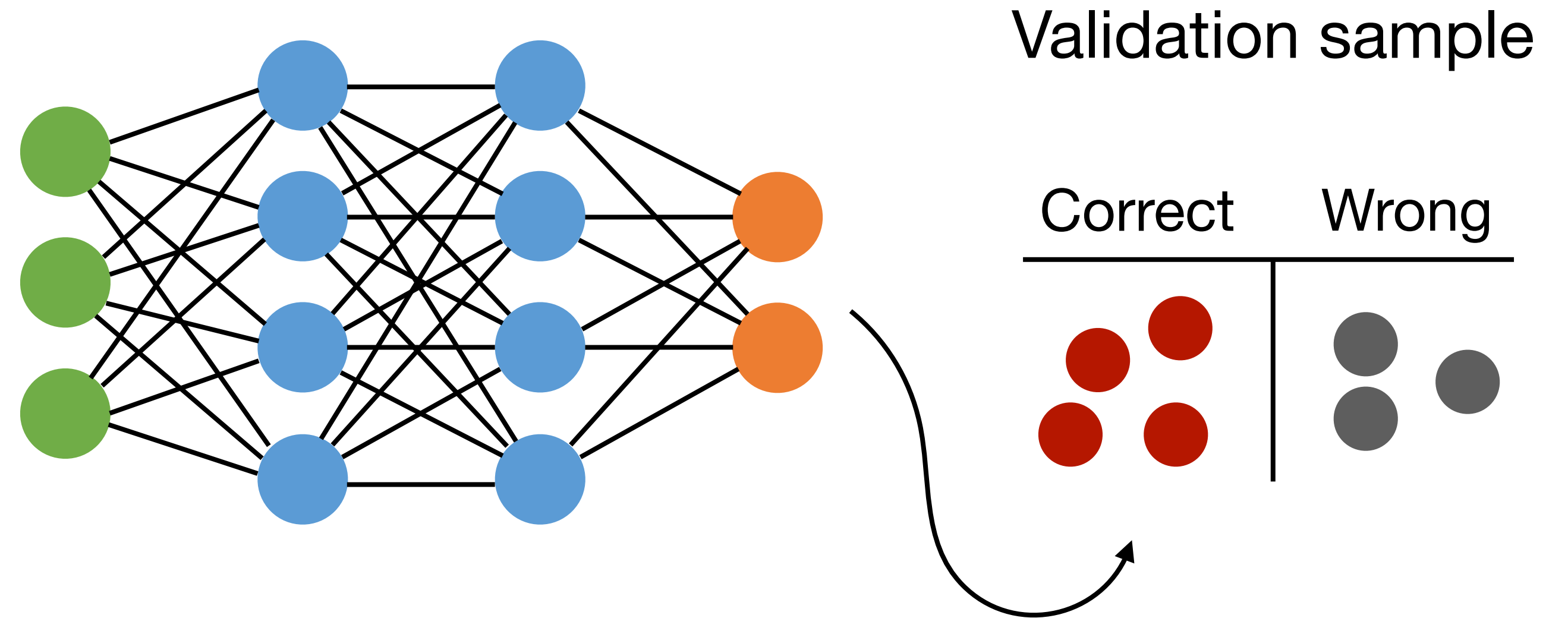
Uncertainty and Clustering

- The simplest way to combine
- Using Uncertainty Sampling, we find texts near the separating curve.
- Clustering all these texts.
- Selecting texts from each cluster.



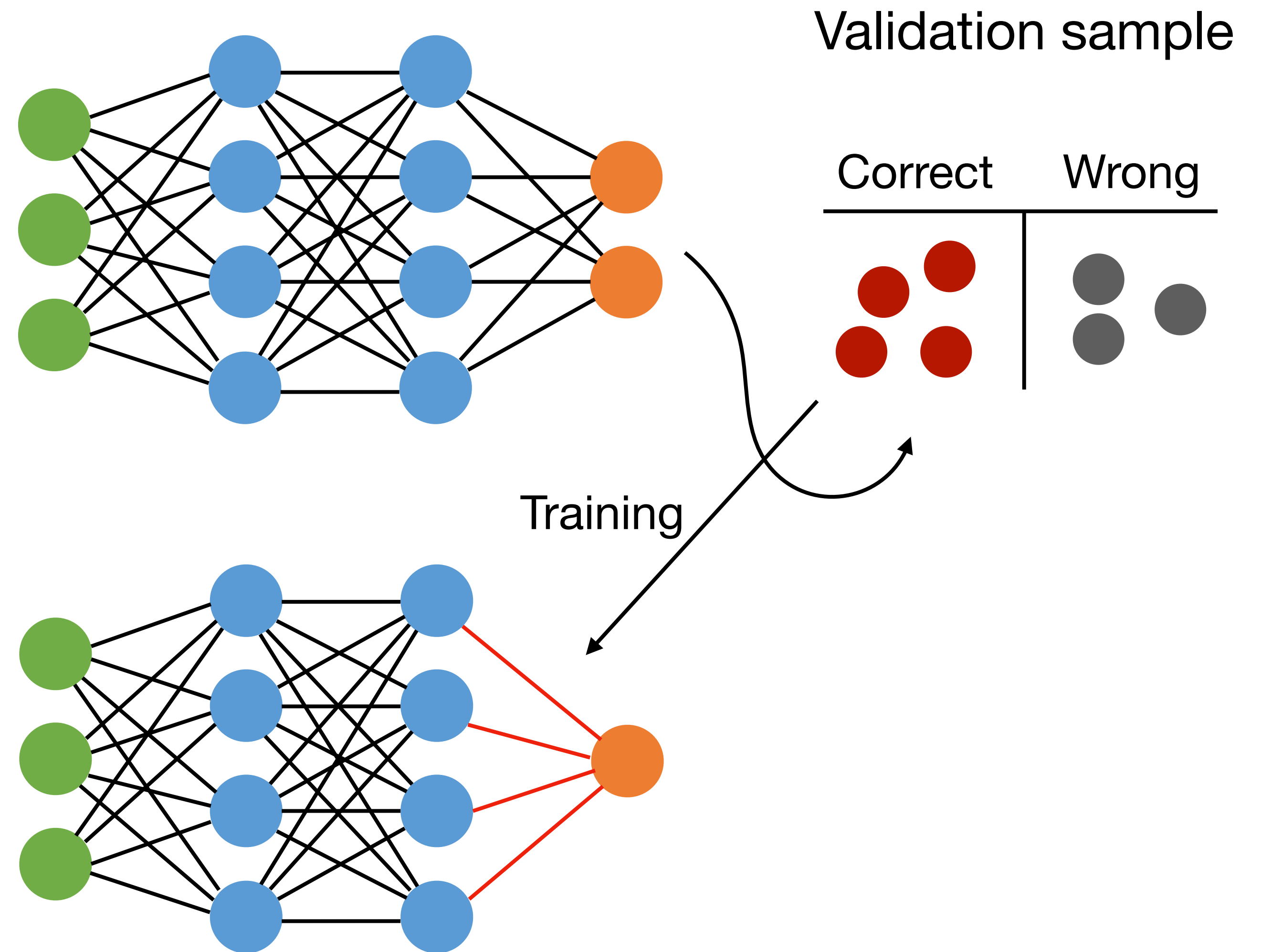
Active transfer learning

- We predict the answers for the validation sample and record the accuracy



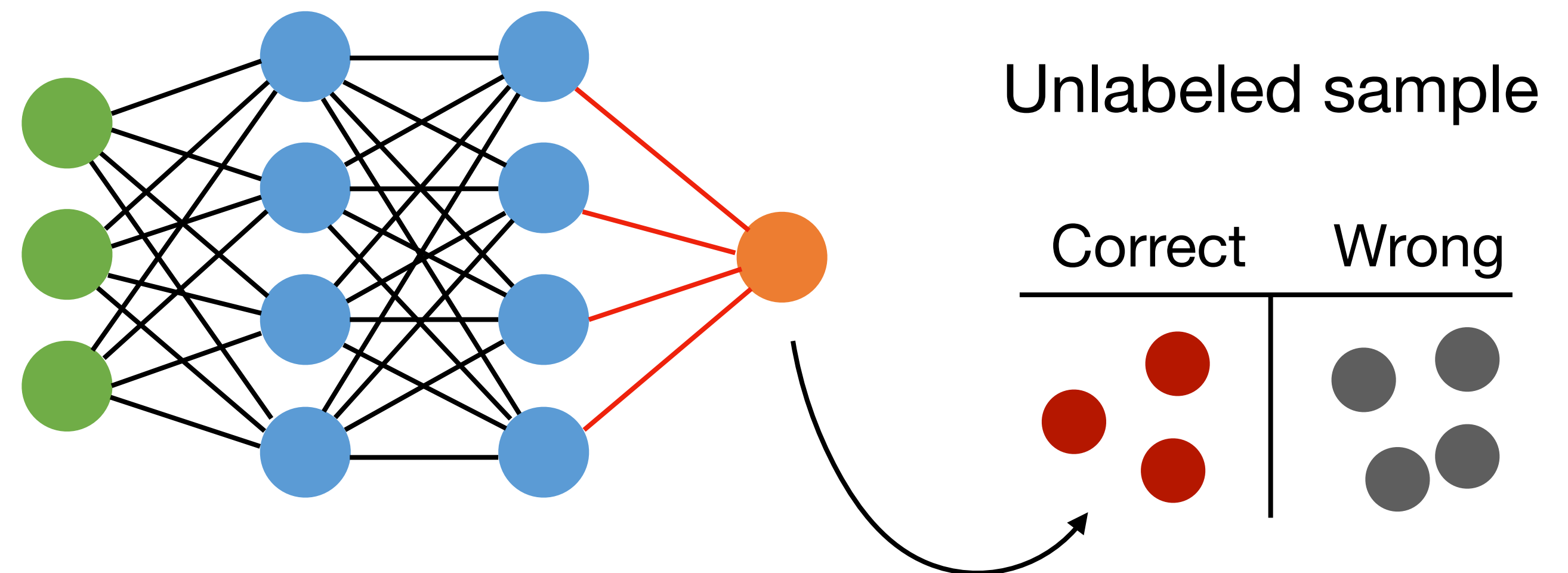
Active transfer learning

- We predict the answers for the validation sample and record the accuracy
- Train the head to predict correctness based on this dataset.



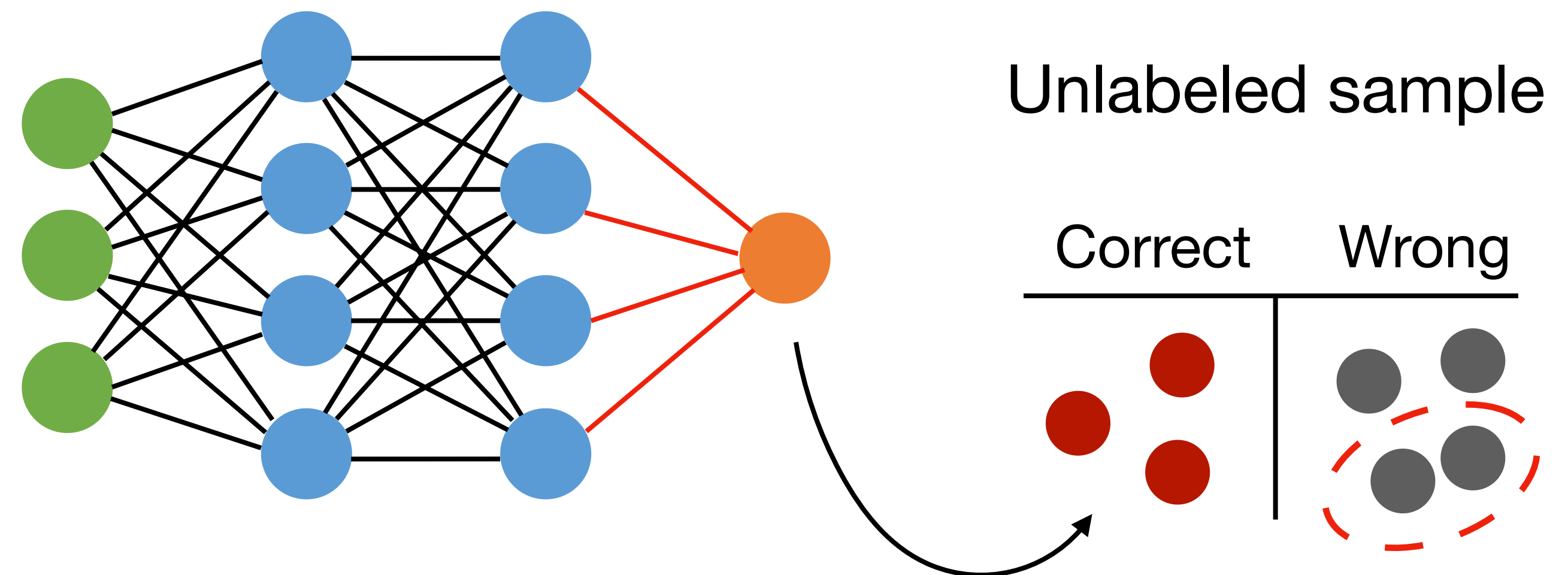
Active transfer learning

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- Predicting accuracy for an unlabeled sample



Active transfer learning

- We predict the answers for the validation sample and record the accuracy
- Train the head to predict correctness based on this dataset.
- Predicting accuracy for an unlabeled sample
- We take texts that are most likely to be incorrectly predicted



Active transfer learning



There's no need to configure any parameters; the model does everything for us.



We don't need a lot of data, since we're only training the head.



Training is very fast.



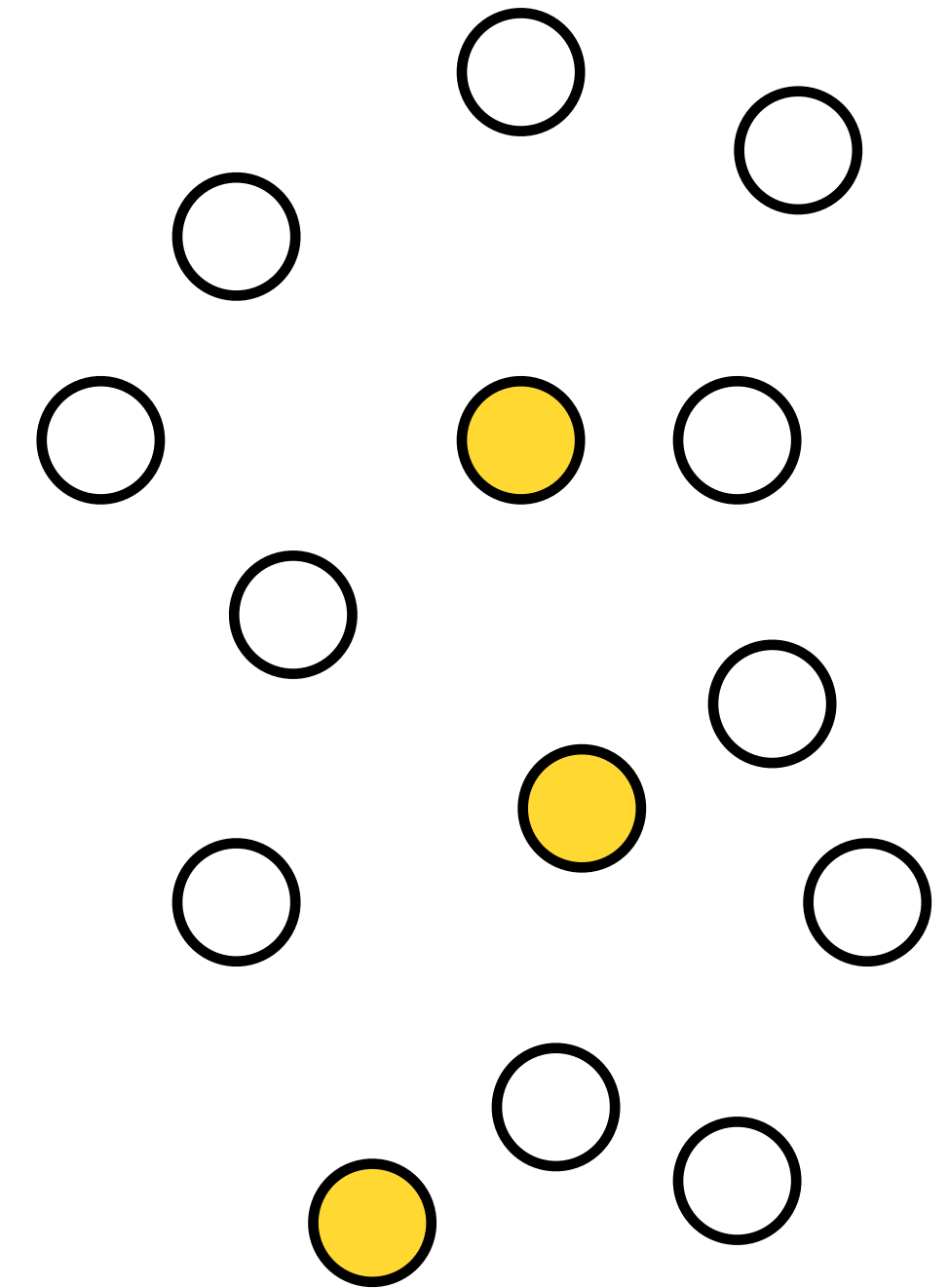
Texts aren't diverse enough.



The model can overfit if the validation set is too small.

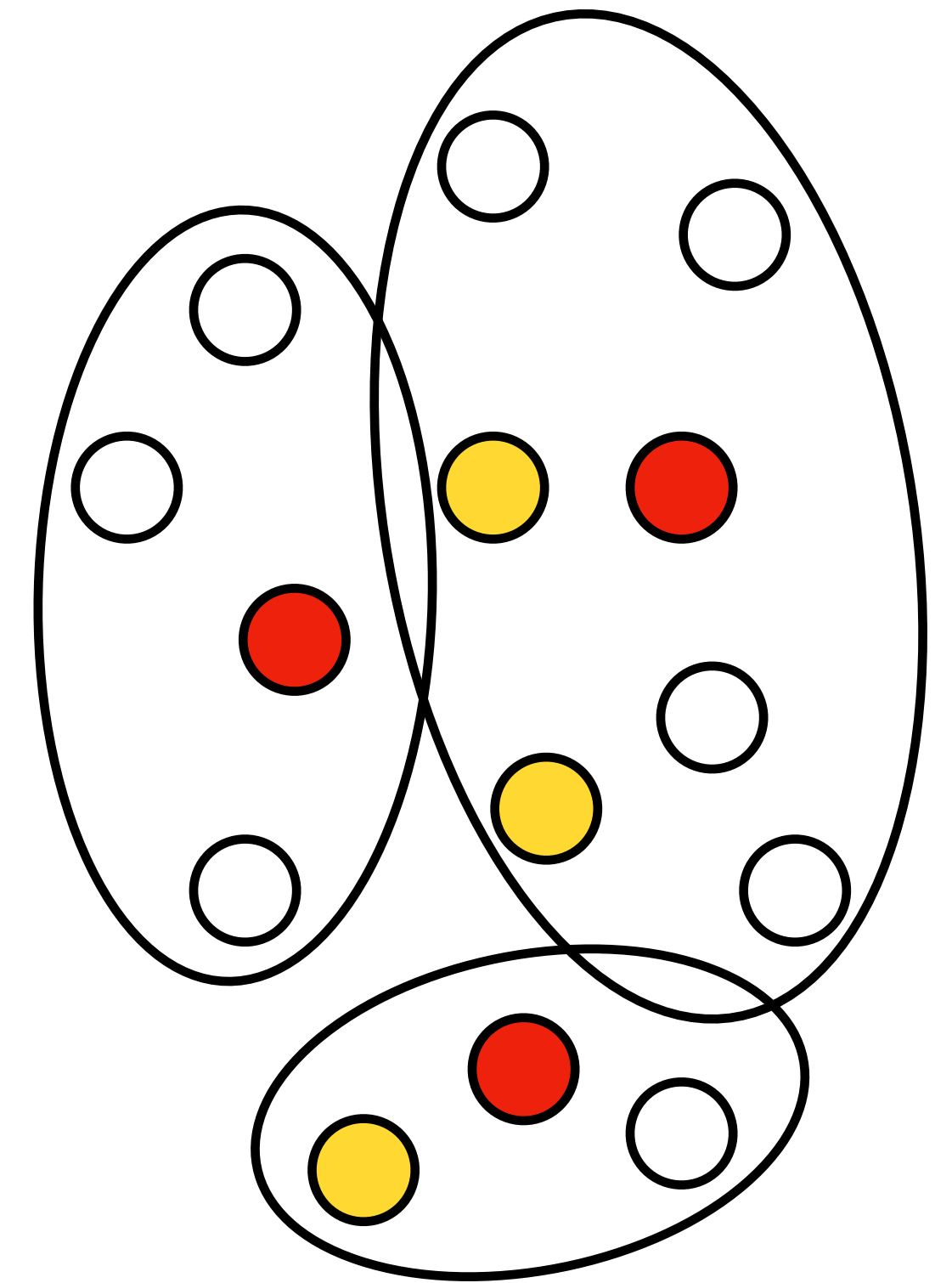
What texts should be labeled at the start?

- The most common case is random sampling.
- This way, we have no control over anything.



What texts should be labeled at the start?

- The most common case is random sampling.
- This way, we have no control over anything.
- It's better to sample cluster centers
- We get good, diverse samples



When to stop active learning?

- The easiest way is to stop when your budget is exhausted.
- You can stop when a certain quality is reached.
 - To do this, you need a large test dataset.  might be expensive
 - Or use cross-validation.  the labeled data is biased
- You can stop when the model becomes confident.  you should normalize the confidence values
- It is possible to measure the difference in models between iterations

Estimating the cost of text markup

- The cost of each text can be assumed to be the same.
 - Texts have different lengths and complexity.
- Cost can be estimated by the number of tokens.
 - This works poorly, as complexity doesn't always depend on length.
- It's better to train a simple model to predict complexity based on various text statistics and the expertise of the annotators.
- Predicting the cost of annotating all texts can be difficult, as AL selects the most complex texts.

How to use labeling cost?

- Do not label up texts above a certain threshold
- You can estimate the contribution from text markup
- it increases as model confidence decreases
- Divide the contribution by the cost (return-on-investment)

$$ROI(x) = \frac{uncertainty(x)}{cost(x)}$$

- Take texts with the highest ROI
- Here, it's useful to normalize the uncertainty value

Acceleration of the cycle

- Retrain the model at each iteration, rather than training it from scratch.
- Train a smaller model, because uncertainty, not accuracy, is important.
- Train the model while the annotators are busy with annotation.
- The dataset will be out of date, but that's not such a big deal.